INNOPOLIS UNIVERSITY

SNE MASTER
TRACK 2026

COURSE: [S26] Computer Forensics and Incident Response

CCF Lab 1 Report

On

Data Acquisition

Student Name: HASIB AL TAHSIN

Application ID: 187623
Student ID: 24

Email: h.tahsin@innopolis.university, hasibjust10@gmail.com

Submission Date: 24.03.2026

Task 1 - Setting up your environment

You need to prepare 2 USB drives. The first one should have a CAINE live environment that will be used to collect evidence. On the second one (this drive will be called drive A) you should deploy that disk image. The disk image is compressed with special utility that preserves original bits intact. You should uncompress it (using FTK imager) and burn it on the flash drive (do not forget about unallocated space).

Solution:

Tools Used

FTK Imager: This software is used to convert the *.E01* image file to *Raw* (*.dd*) format.

Rufus: This tool is used to write the Raw image file *bit-by-bit* to the USB drive.

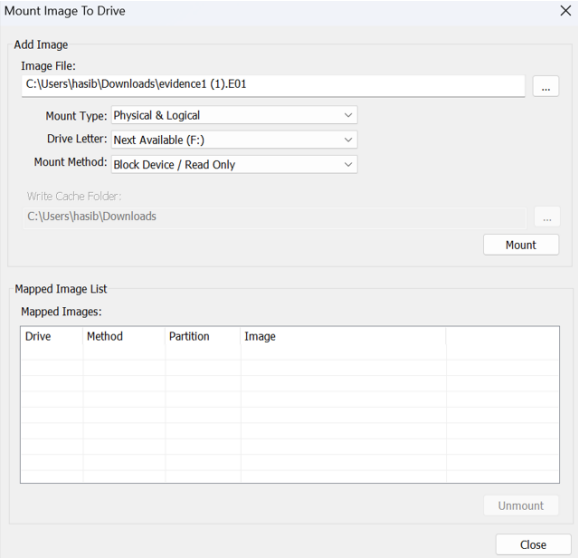
VMware Workstation: This is used to run the CAINE live environment.

Windows Disk Management: This is used to verify the partitioning and unallocated space of the drive.

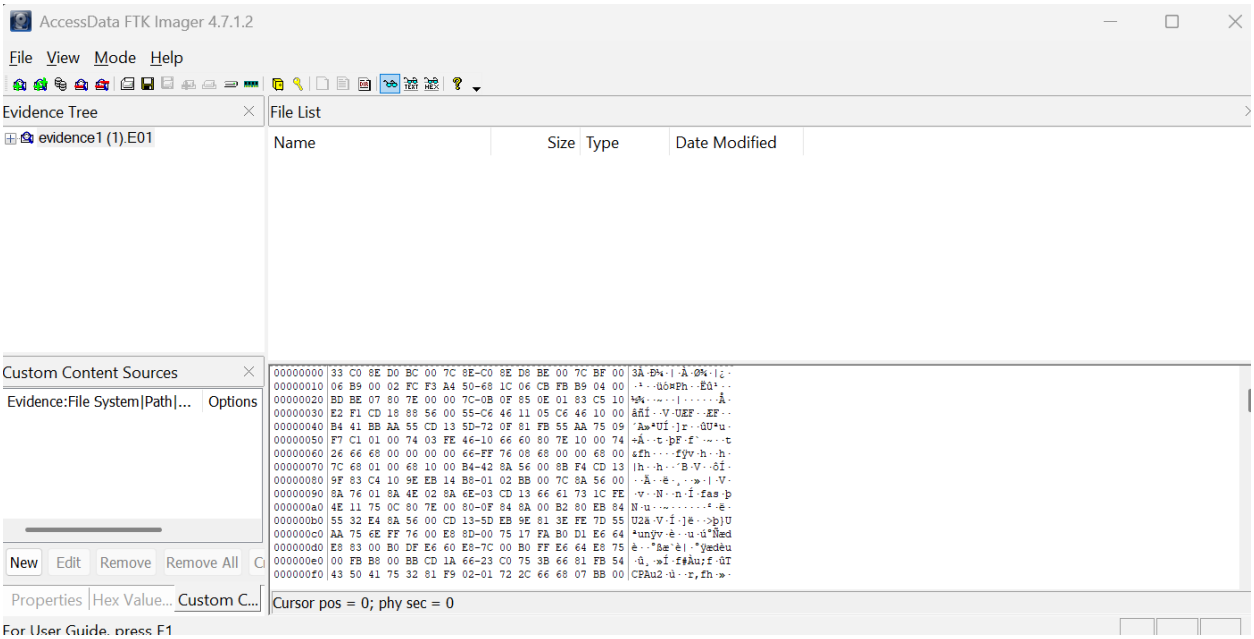
Procedures:

Drive A Preparation (Evidence Deployment):

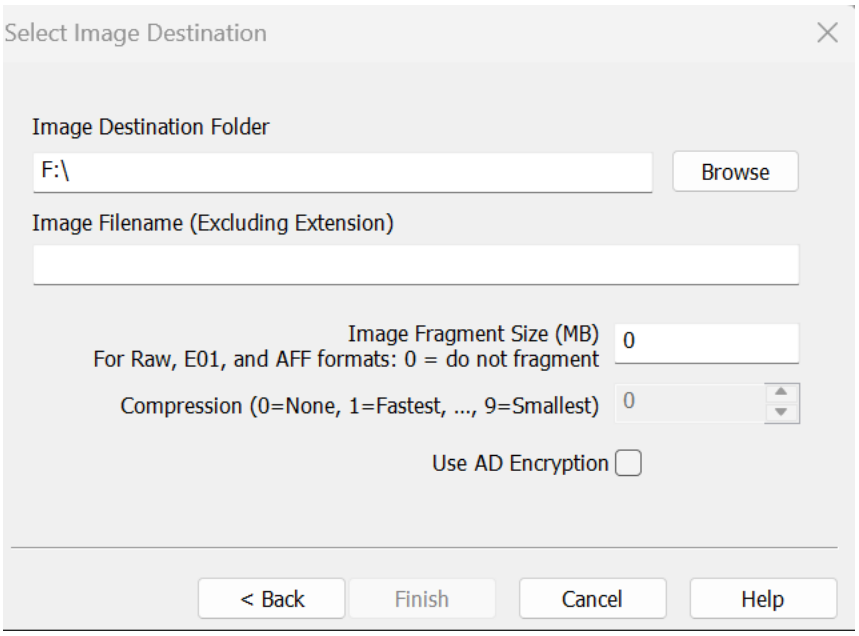
Uncompress the image: The *evidence1 (1).E01* file is loaded as a source in *FTK Imager*.

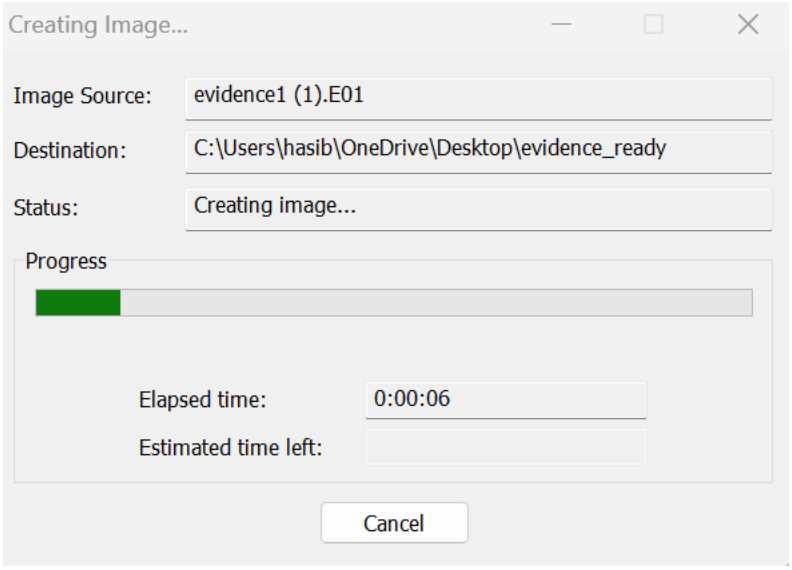


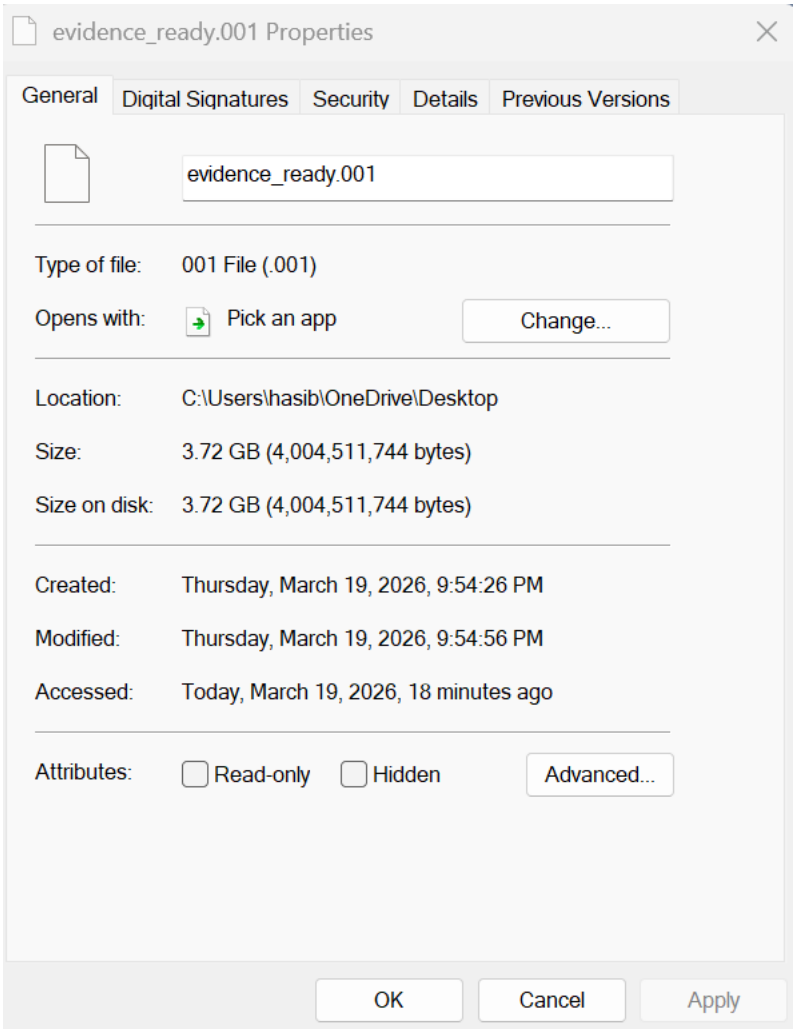
The image was then exported in *Raw (dd)* format, preserving the original bits.



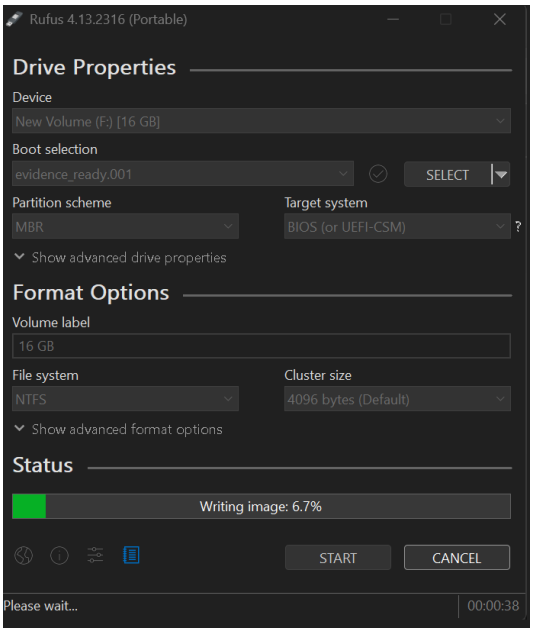
While exporting, the *Fragment Size* was set to 0, so that the entire *3.72 GB* of data is available in a single file. As a result, I got the file *evidence_ready.001.*



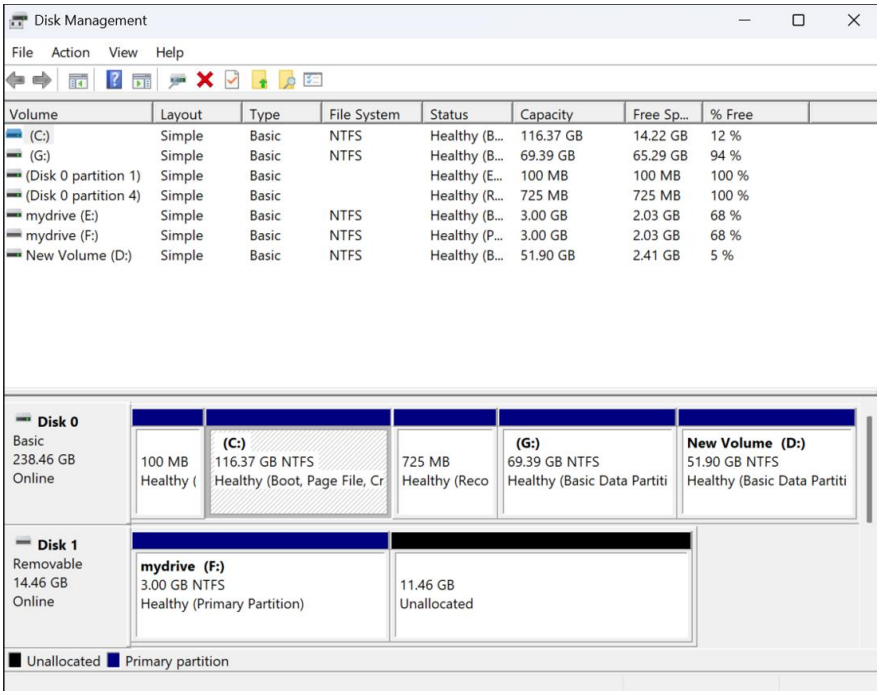

Creating Image...
Image Source:
evidence1 (1).E01
Destination:
C:\Users\hasib\OneDrive\Desktop\evidence_ready
Status:
Creating image...
Progress
Elapsed time:
0:00:06
Estimated time left:
Cancel


evidence_ready.001 Properties
General
Digital Signatures
Security
Details
Previous Versions
evidence_ready.001
Type of file:
001 File (.001)
Opens with:
Pick an app
Change...
Location:
C:\Users\hasib\OneDrive\Desktop
Size:
3.72 GB (4,004,511,744 bytes)
Size on disk:
3.72 GB (4,004,511,744 bytes)
Created:
Thursday, March 19, 2026, 9:54:26 PM
Modified:
Thursday, March 19, 2026, 9:54:56 PM
Accessed:
Today, March 19, 2026, 18 minutes ago
Attributes:
Read-only
Hidden
Advanced...
OK
Cancel
Apply

Burning with Rufus: Connect the 16GB flash drive to the PC and open Rufus. Then select the *'.001'* file we created in the boot selection. Rufus automatically selects *"DD Image Mode"*. This ensures that the image is written to the very beginning of the drive and the rest is left blank.

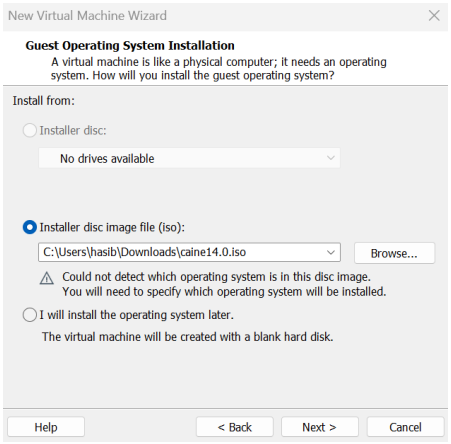


Result Verification: Windows *Disk Management* shows that a 3 GB partition has been created at the beginning of the pen drive and the rest is *Unallocated.* This was the main condition of the lab.

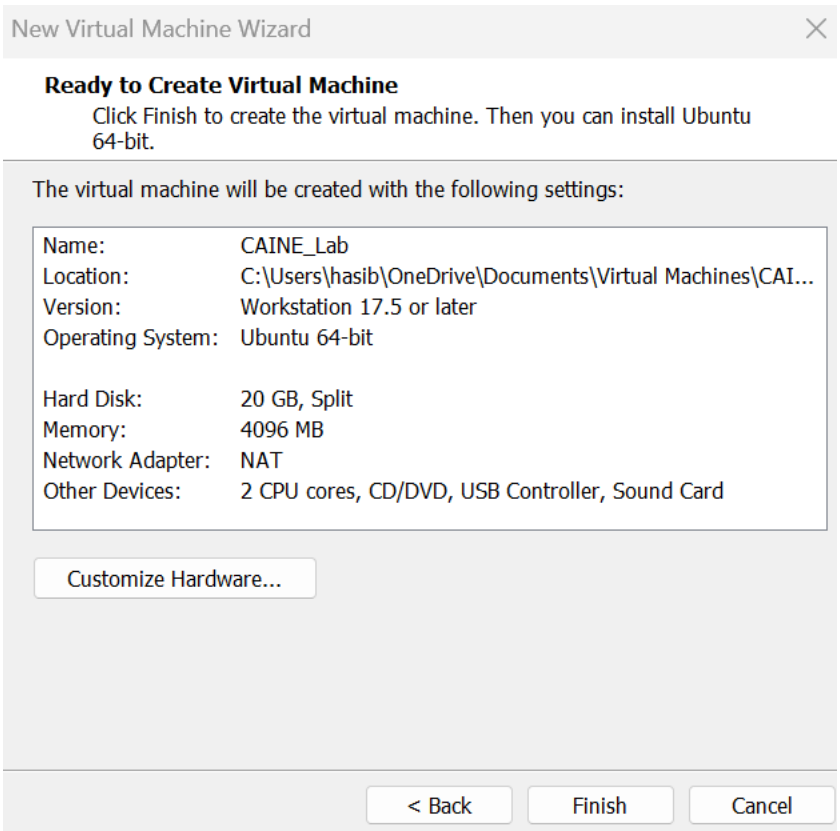


CAINE Live Environment Setup in VMware:

Creating Virtual Machine: A new VM is created in VMware and the *CAINE ISO* file is mounted as the operating system.



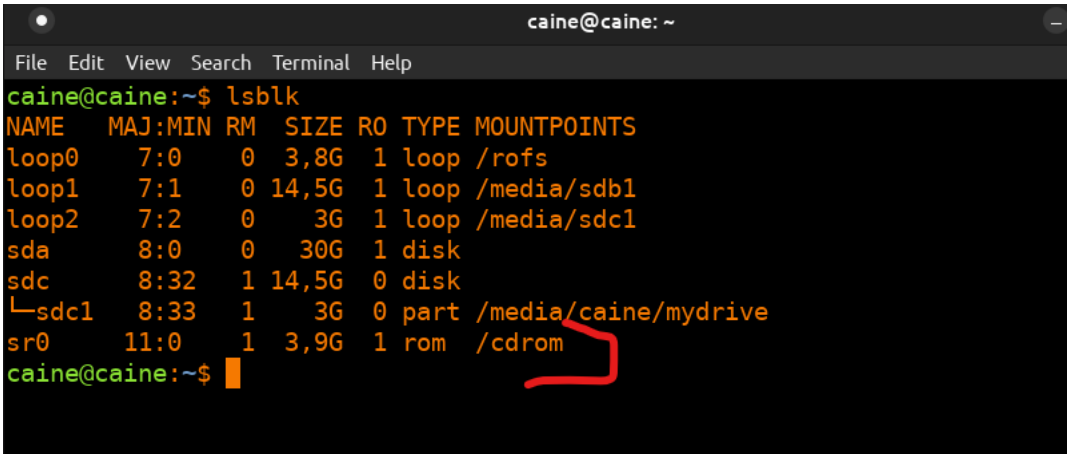
Sufficient RAM and processor are allocated so that the forensic tools run smoothly.



Connecting the drive: Then, *Drive A (Flash Drive)* prepared from VMware's *Removable Devices* option and an empty Evidence Storage drive are connected to the virtual machine.

Starting CAINE: After booting the VM, the graphical interface of CAINE is opened.

Then, using *Terminal* or *Disk Utility*, it is ensured that *Drive A* is in *Read-Only* or *Blocked* state so that the evidence is not changed. In this entire process, no changes are made to the hash value or bit level of the original image.



Task 2 - Imaging

1. Discuss how you can retrieve an image from an, currently off-line, USB stick in a forensically sound manner. Create and describe this method.

Solution:

To ensure that data is not altered in any way while collecting data from an offline USB stick, the following procedure is followed:

- **Write Blocking:** First, ensure that the drive is in *Read-Only mode.* In CAINE, this is ensured through a software level write-blocker or Mounter tool.
- **Bit-by-Bit Copy:** Instead of a simple *'Copy-Paste'*, a forensic tool copies every bit of the drive (even deleted data).
- **Hashing:** Calculate the MD5 or SHA-1 hash value of the drive before and after creating the image. If the hash is the same both times, it will be proven that the evidence has not been altered during the investigation.

- **Documentation:** Keep a note of the timestamp of each step and the version of the tool used.

2. Write a one-line description, or note a useful feature for the following tools included in CAINE: Guymager, Disk Image Mounter, dcfldd / dc3dd, kpartx.

Solution:

Guymager: This is a graphical forensic imager tool that works very fast and it automatically generates hash verification reports.

Disk Image Mounter: This is a utility that mounts forensic image files (such as *.dd, .E01*) as drives on the system and allows you to view the files inside.

dcfldd / dc3dd: These are advanced versions of the dd command, which can show real-time hashing and progress bars while creating the image.

kpartx: This tool helps to identify and mount different partitions inside an image file.

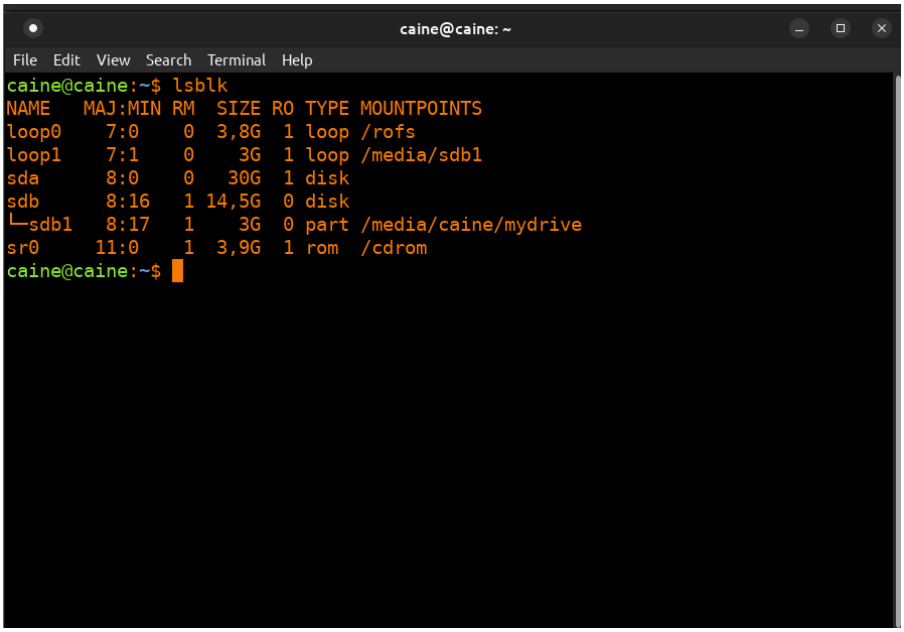
3. Follow your method to retrieve the image from drive A. Please use timestamps, explain every tool and note down the version. For the purpose of speed. Make sure both team members have access to the retrieved image. You can use your PCs as an evidence sharing platform.

Solution:

Using the information from the collected *.info file*, these steps were followed:

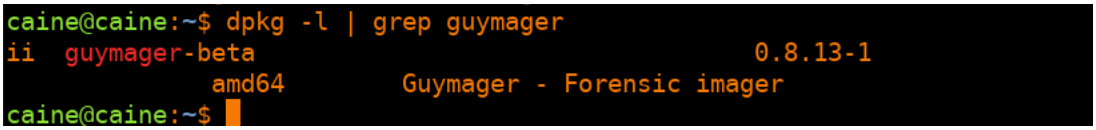
1. Drive Identification

- **Time:** 21:06:07
- **Tool:** The Kingston USB drive was identified as */dev/sdb* using the *lsblk* command.
- **Version:** *lsblk* from util-*linux 2.37.2* (CAINE default).
- **Result:** I have seen my drive as */dev/sdc or /dev/sdd* and its partition is mounted at */media/caine/mydrive.*



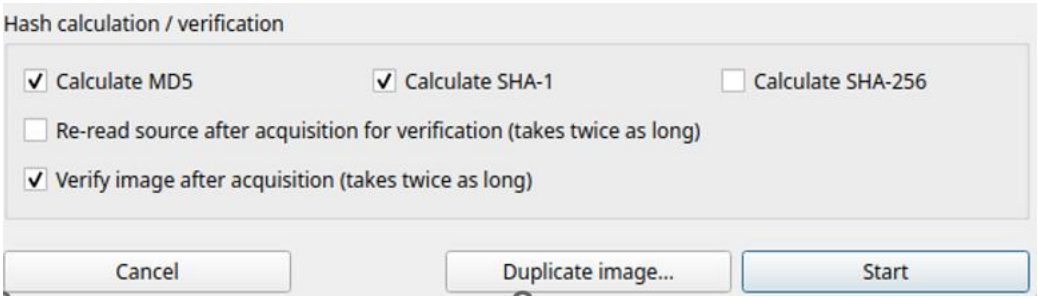
2. Forensic Imaging Process

- **Tool:** Guymager
- **Version:** 0.8.13-1
- **Explanation:** This is a graphical imager that creates *bit-by-bit* copies. I used the Expert Witness Format (*.E01*) format because it saves metadata and hashes within the image.



3: Start image creation

- **Task:** Right-clicked on the drive in *Guymager* and selected "*Acquire Image".*
- **Settings: Format**: I selected *Expert Witness Format (.E01).*
- **Destination:** I selected a folder on the PC where the image was saved.
- **Verification:** I checked the MD5 and SHA-1 *boxes.*



8. Verification & Integrity

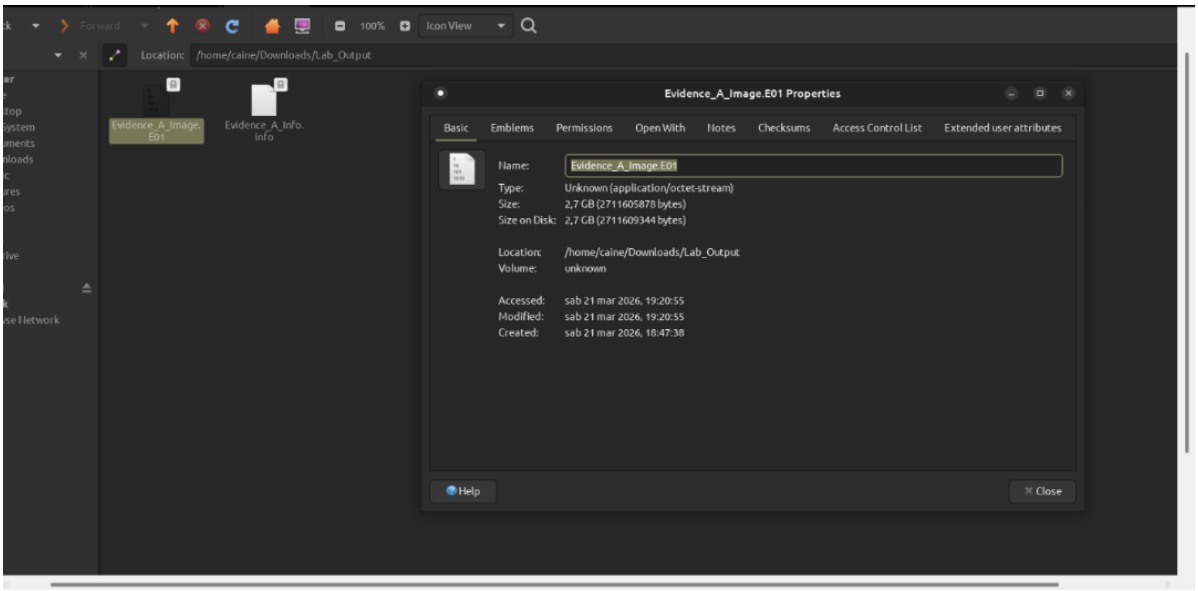
- **Time:** 21:25:08
- **Method:** MD5 and SHA-1 hash calculation.
- **Result:** MD5 Hash Verified (*b5db053974301da99367027c912433f8*).
- **Conclusion:** The hashes of the source drive and the image match exactly, proving that data integrity is maintained (No data altered).

```
 98 MD5 hash                   : 7df609869e8c3b9cf9b163f8b755bd6c
 99 MD5 hash verified source   : --
100 MD5 hash verified image    : 7df609869e8c3b9cf9b163f8b755bd6c
101 SHA1 hash                  : 5e23da7bc81923c0e2a67a2b95da97c3e6ce91d7
102 SHA1 hash verified source  : --
103 SHA1 hash verified image   : 5e23da7bc81923c0e2a67a2b95da97c3e6ce91d7
104 SHA256 hash                : --
105 SHA256 hash verified source: --
106 SHA256 hash verified image : --
107 Image verification OK. The image contains exactly the data that was written.
```

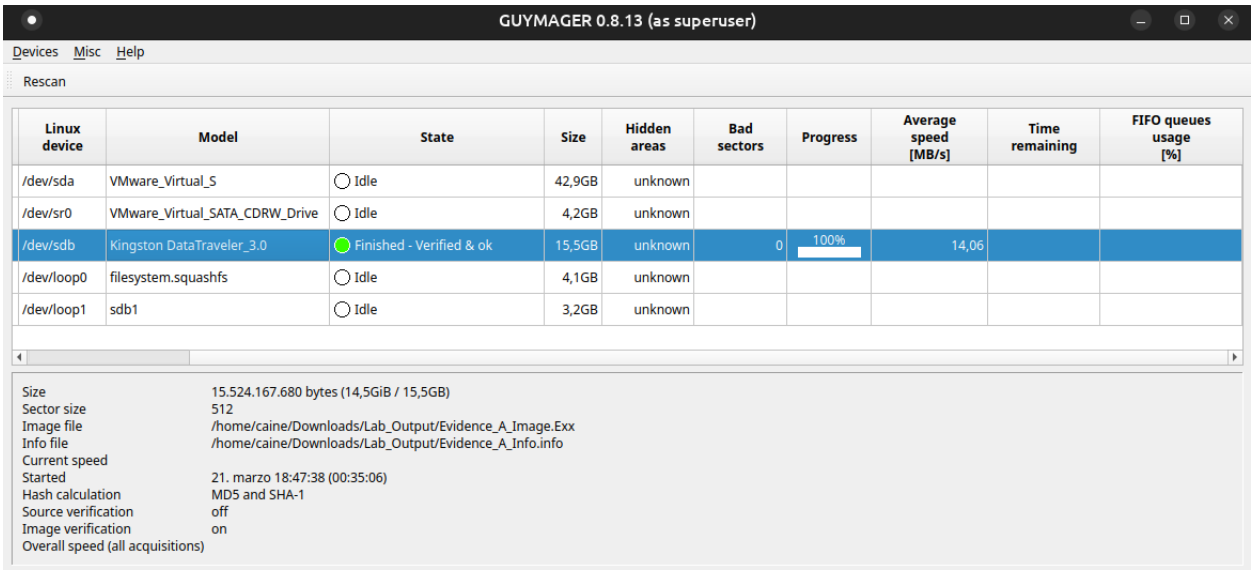
8. Evidence Sharing

- **Method:** VMware Shared Folders or Host-Guest drag and drop.
- **Action:** The *.E01* and *.info* files from the */home/caine/Downloads/Lab_Output/* folder were copied to the host PC (*Windows*).
- **Team Access:** The files from the host PC were shared with other members via Google Drive and a shared local drive so that both of us could analyze.

.info File:



Guymager's work is finished with "*Finished - Verified & ok*".



4. Read about CAINE Linux and its features while waiting on the dump to finish.

1. Why would you use a Forensic distribution and what are the main differences between a regular distribution?

2. When would you use a live environment and when would you use an installed environment?

3. What are the policies of CAINE?

Solution:

1. Reasons to use: Forensic distributions (e.g. *CAINE*) are designed in such a way that people do not alter any evidence or data on the drive. *Regular Linux* (e.g. *Ubuntu* or *Windows*) writes some files to the drive or changes the timestamp when a drive is connected, which is not forensically acceptable.

Regular Distribution vs Forensic Distribution

The main difference between a regular distribution (*Regular OS*) and a forensic distribution (*Forensic OS*) in digital forensics is the Auto-Mount process. In a regular OS, when a drive is connected, it is automatically mounted and data can be written, but in a forensic OS, the drive is not auto-mounted and is *Read-Only* by default. A regular OS is

mainly equipped with entertainment or office tools, whereas a forensic OS only has data recovery, decryption, and analysis tools. In addition, in a normal OS, the metadata of the file (such as *Access Time*) may change, but the forensic environment maintains the data integrity or original state of the drive.

2. Live Environment vs. Installed Environment

Live Environment: When I want to collect data directly from a suspicious PC or server *(Live Acquisition*), use Live mode. It does not leave any traces on the PC's hard drive.

Installed Environment: When I have images already collected in the laboratory (*.E01 file*) and need to perform complex analysis over a long period of time, then the installed environment on the PC is convenient.

3. CAINE's main policies

- Non-Invasive Policy: It will not make any changes to the system.
- **Write-Blocking Policy:** By default, all drives are mounted in *Read-Only* mode.
- **Evidence Integrity:** Maintaining the chain of custody and hash value of evidence.

5. As soon as your dump finishes, start a tool to create a timeline on the image. You will need this timeline later in the assignment. Hints: log2timeline.py.

Solution:

After the imaging is complete, a Timeline is created. A timeline is a sequential list of when a file was created or deleted within the evidence.

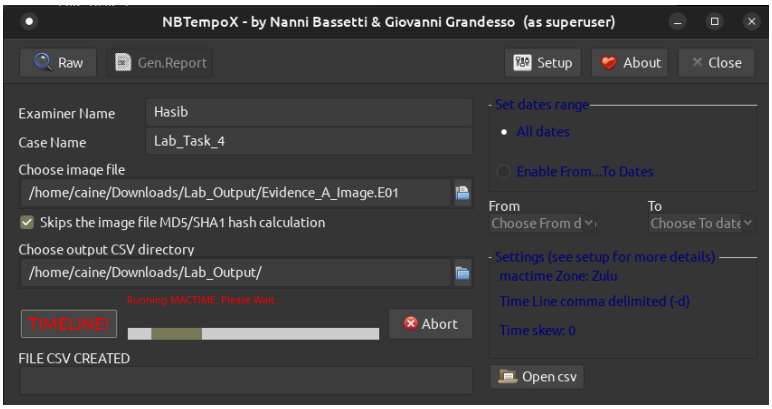
Initial Attempt & Troubleshooting:

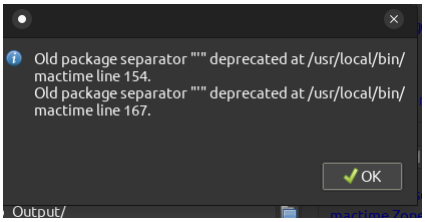
As per the assignment hint, an initial attempt was made to create a timeline using the *log2timeline.py* tool.

```
caine@caine:~/Desktop$ log2timeline timeline.plaso /home/caine/Downloads/Lab_Output/Evidence_A_Image.E01
log2timeline: command not found
```

The tool was not installed by default in the *CAINE Linux* (Kernel 6.8.0-52-generic) environment *(Command not found error).* An attempt to install the package also failed *(Unable to locate package python3-plaso).*

GUI Attempt: The graphical tool *NBTempoX* was then used.

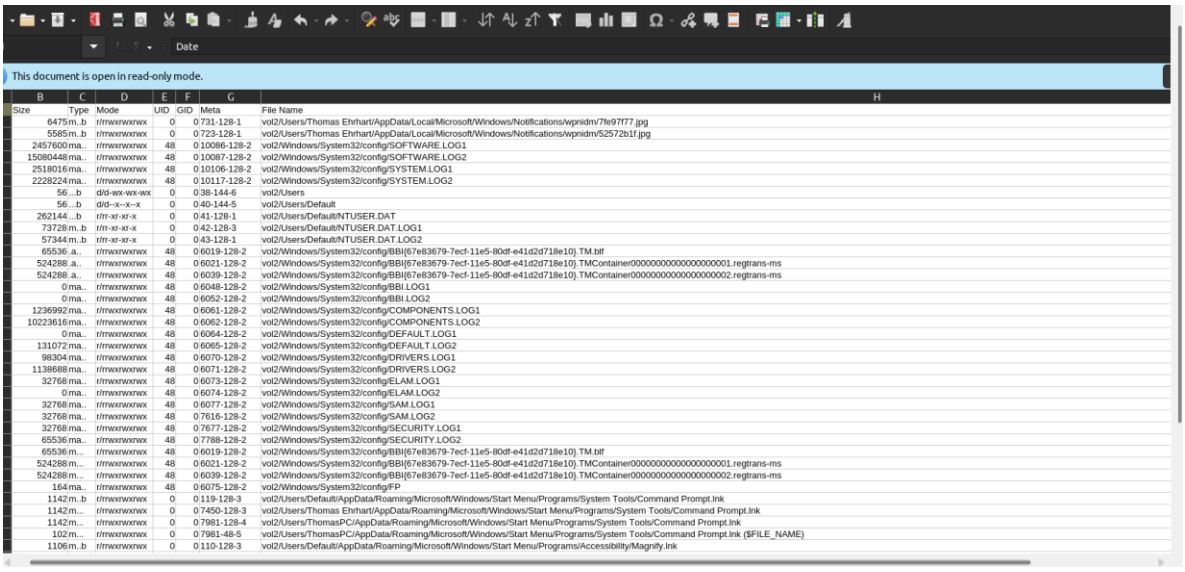




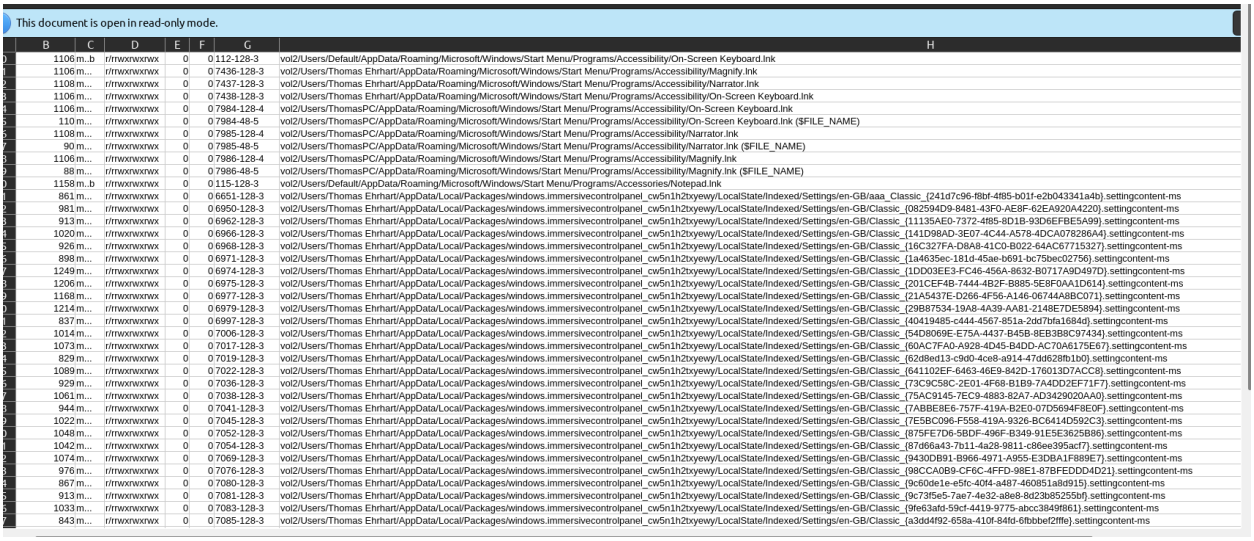
Checking the timeline:

Timeline creation started at *21/03/2026* to extract artifact metadata from *Drive A* image.

After opening the CSV file:

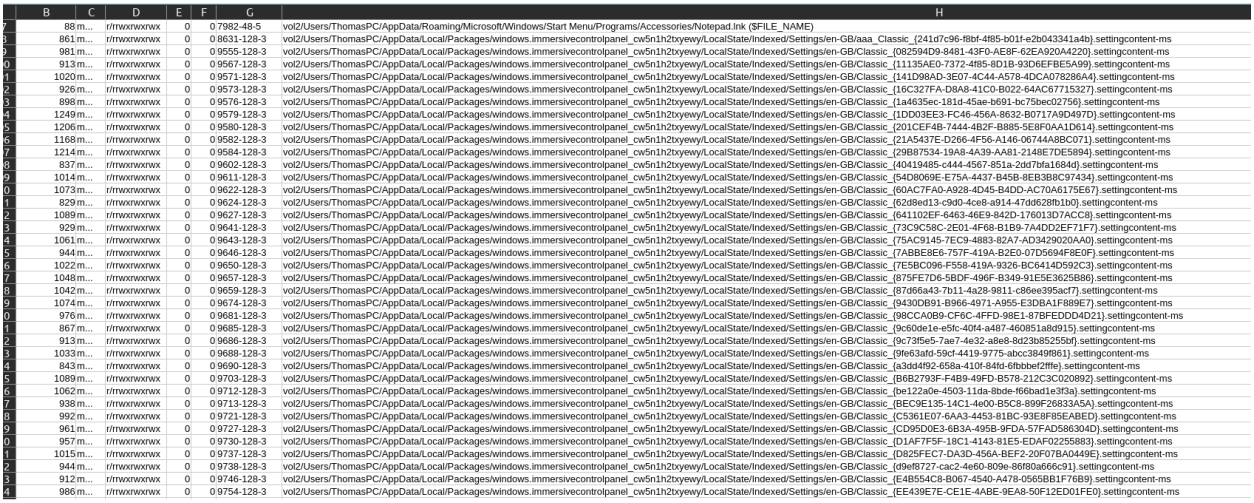


Size	Type	Mode	UID	GID	Meta	File Name
6475	m..b	r/rwxrwxrwx	0	0	731-128-1	vol2/Users/Thomas Ehrhart/AppData/Local/Microsoft/Windows/Notifications/wpnidm/7fe97f77.jpg
5585	m..b	r/rwxrwxrwx	0	0	723-128-1	vol2/Users/Thomas Ehrhart/AppData/Local/Microsoft/Windows/Notifications/wpnidm/52572b1f.jpg
2457600	ma..	r/rwxrwxrwx	48	0	10086-128-2	vol2/Windows/System32/config/SOFTWARE.LOG1
15080448	ma..	r/rwxrwxrwx	48	0	10087-128-2	vol2/Windows/System32/config/SOFTWARE.LOG2
2518016	ma..	r/rwxrwxrwx	48	0	10106-128-2	vol2/Windows/System32/config/SYSTEM.LOG1
2228224	ma..	r/rwxrwxrwx	48	0	10117-128-2	vol2/Windows/System32/config/SYSTEM.LOG2
56	...b	d/d-wx-wx-wx	0	0	38-144-6	vol2/Users
56	...b	d/d--x--x--x	0	0	40-144-5	vol2/Users/Default
262144	...b	r/rr-xr-xr-x	0	0	41-128-1	vol2/Users/Default/NTUSER.DAT
73728	m..b	r/rr-xr-xr-x	0	0	42-128-3	vol2/Users/Default/NTUSER.DAT.LOG1
57344	m..b	r/rr-xr-xr-x	0	0	43-128-1	vol2/Users/Default/NTUSER.DAT.LOG2
65536	.a..	r/rwxrwxrwx	48	0	6019-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TM.blf
524288	.a..	r/rwxrwxrwx	48	0	6021-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000001.regtrans-ms
524288	.a..	r/rwxrwxrwx	48	0	6039-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000002.regtrans-ms
0	ma..	r/rwxrwxrwx	48	0	6048-128-2	vol2/Windows/System32/config/BBI.LOG1
0	ma..	r/rwxrwxrwx	48	0	6052-128-2	vol2/Windows/System32/config/BBI.LOG2
1236992	ma..	r/rwxrwxrwx	48	0	6061-128-2	vol2/Windows/System32/config/COMPONENTS.LOG1
10223616	ma..	r/rwxrwxrwx	48	0	6062-128-2	vol2/Windows/System32/config/COMPONENTS.LOG2
0	ma..	r/rwxrwxrwx	48	0	6064-128-2	vol2/Windows/System32/config/DEFAULT.LOG1
131072	ma..	r/rwxrwxrwx	48	0	6065-128-2	vol2/Windows/System32/config/DEFAULT.LOG2
98304	ma..	r/rwxrwxrwx	48	0	6070-128-2	vol2/Windows/System32/config/DRIVERS.LOG1
1138688	ma..	r/rwxrwxrwx	48	0	6071-128-2	vol2/Windows/System32/config/DRIVERS.LOG2
32768	ma..	r/rwxrwxrwx	48	0	6073-128-2	vol2/Windows/System32/config/ELAM.LOG1
0	ma..	r/rwxrwxrwx	48	0	6074-128-2	vol2/Windows/System32/config/ELAM.LOG2
32768	ma..	r/rwxrwxrwx	48	0	6077-128-2	vol2/Windows/System32/config/SAM.LOG1
32768	ma..	r/rwxrwxrwx	48	0	7616-128-2	vol2/Windows/System32/config/SAM.LOG2
32768	ma..	r/rwxrwxrwx	48	0	7677-128-2	vol2/Windows/System32/config/SECURITY.LOG1
65536	ma..	r/rwxrwxrwx	48	0	7788-128-2	vol2/Windows/System32/config/SECURITY.LOG2
65536	m...	r/rwxrwxrwx	48	0	6019-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TM.blf
524288	m...	r/rwxrwxrwx	48	0	6021-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000001.regtrans-ms
524288	m...	r/rwxrwxrwx	48	0	6039-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000002.regtrans-ms
164	ma..	r/rwxrwxrwx	48	0	6075-128-2	vol2/Windows/System32/config/FP
1142	m..b	r/rwxrwxrwx	0	0	119-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk
1142	m...	r/rwxrwxrwx	0	0	7450-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk
1142	m...	r/rwxrwxrwx	0	0	7981-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk
102	m...	r/rwxrwxrwx	0	0	7981-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk ($FILE_NAME)
1106	m..b	r/rwxrwxrwx	0	0	110-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk



This document is open in read-only mode.

B	C	D	E	F	G	H
1106	m..b	r/rwxrwxrwx	0	0	112-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk
1106	m...	r/rwxrwxrwx	0	0	7436-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk
1108	m...	r/rwxrwxrwx	0	0	7437-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Narrator.lnk
1106	m...	r/rwxrwxrwx	0	0	7438-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk
1106	m...	r/rwxrwxrwx	0	0	7984-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk
110	m...	r/rwxrwxrwx	0	0	7984-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk ($FILE_NAME)
1108	m...	r/rwxrwxrwx	0	0	7985-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Narrator.lnk
90	m...	r/rwxrwxrwx	0	0	7985-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Narrator.lnk ($FILE_NAME)
1106	m...	r/rwxrwxrwx	0	0	7986-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk
88	m...	r/rwxrwxrwx	0	0	7986-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk ($FILE_NAME)
1158	m..b	r/rwxrwxrwx	0	0	115-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessories/Notepad.lnk
861	m...	r/rwxrwxrwx	0	0	6651-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/aaa_Classic_{241d7c96-f8bf-4f85-b01f-e2b043341a4b}.settingcontent-ms
981	m...	r/rwxrwxrwx	0	0	6950-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{082594D9-8481-43F0-AE8F-62EA920A4220}.settingcontent-ms
913	m...	r/rwxrwxrwx	0	0	6962-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{11135AE0-7372-4f85-8D1B-93D6EFBE5A99}.settingcontent-ms
1020	m...	r/rwxrwxrwx	0	0	6966-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{141D98AD-3E07-4C44-A578-4DCA078286A4}.settingcontent-ms
926	m...	r/rwxrwxrwx	0	0	6968-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{16C327FA-D8A8-41C0-B022-64AC67715327}.settingcontent-ms
898	m...	r/rwxrwxrwx	0	0	6971-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1a4635ec-181d-45ae-b691-bc75bec02756}.settingcontent-ms
1249	m...	r/rwxrwxrwx	0	0	6974-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1DD03EE3-FC46-456A-8632-B0717A9D497D}.settingcontent-ms
1206	m...	r/rwxrwxrwx	0	0	6975-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{201CEF4B-7444-4B2F-B885-5E8F0AA1D614}.settingcontent-ms
1168	m...	r/rwxrwxrwx	0	0	6977-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{21A5437E-D266-4F56-A146-06744A8BC071}.settingcontent-ms
1214	m...	r/rwxrwxrwx	0	0	6979-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{29B87534-19A8-4A39-AA81-2148E7DE5894}.settingcontent-ms
837	m...	r/rwxrwxrwx	0	0	6997-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{40419485-c444-4567-851a-2dd7bfa1684d}.settingcontent-ms
1014	m...	r/rwxrwxrwx	0	0	7006-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{54D8069E-E75A-4437-B45B-8EB3B8C97434}.settingcontent-ms
1073	m...	r/rwxrwxrwx	0	0	7017-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{60AC7FA0-A928-4D45-B4DD-AC70A6175E67}.settingcontent-ms
829	m...	r/rwxrwxrwx	0	0	7019-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{62d8ed13-c9d0-4ce8-a914-47dd628fb1b0}.settingcontent-ms
1089	m...	r/rwxrwxrwx	0	0	7022-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{641102EF-6463-46E9-842D-176013D7ACC8}.settingcontent-ms
929	m...	r/rwxrwxrwx	0	0	7036-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{73C9C58C-2E01-4F68-B1B9-7A4DD2EF71F7}.settingcontent-ms
1061	m...	r/rwxrwxrwx	0	0	7038-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{75AC9145-7EC9-4883-82A7-AD3429020AA0}.settingcontent-ms
944	m...	r/rwxrwxrwx	0	0	7041-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7ABBE8E6-757F-419A-B2E0-07D5694F8E0F}.settingcontent-ms
1022	m...	r/rwxrwxrwx	0	0	7045-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7E5BC096-F558-419A-9326-BC6414D592C3}.settingcontent-ms
1048	m...	r/rwxrwxrwx	0	0	7052-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{875FE7D6-5BDF-496F-B349-91E5E3625B86}.settingcontent-ms
1042	m...	r/rwxrwxrwx	0	0	7054-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{87d66a43-7b11-4a28-9811-c86ee395acf7}.settingcontent-ms
1074	m...	r/rwxrwxrwx	0	0	7069-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9430DB91-B966-4971-A955-E3DBA1F889E7}.settingcontent-ms
976	m...	r/rwxrwxrwx	0	0	7076-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{98CCA0B9-CF6C-4FFD-98E1-87BFEDDD4D21}.settingcontent-ms
867	m...	r/rwxrwxrwx	0	0	7080-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c60de1e-e5fc-40f4-a487-460851a8d915}.settingcontent-ms
913	m...	r/rwxrwxrwx	0	0	7081-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c73f5e5-7ae7-4e32-a8e8-8d23b85255bf}.settingcontent-ms
1033	m...	r/rwxrwxrwx	0	0	7083-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9fe63afd-59cf-4419-9775-abcc3849f861}.settingcontent-ms
843	m...	r/rwxrwxrwx	0	0	7085-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{a3dd4f92-658a-410f-84fd-6fbbbef2fffe}.settingcontent-ms



B	C	D	E	F	G	H
88	m...	r/rwxrwxrwx	0	0	7982-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessories/Notepad.lnk ($FILE_NAME)
861	m...	r/rwxrwxrwx	0	0	8631-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/aaa_Classic_{241d7c96-f8bf-4f85-b01f-e2b043341a4b}.settingcontent-ms
981	m...	r/rwxrwxrwx	0	0	9555-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{082594D9-8481-43F0-AE8F-62EA920A4220}.settingcontent-ms
913	m...	r/rwxrwxrwx	0	0	9567-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{11135AE0-7372-4f85-8D1B-93D6EFBE5A99}.settingcontent-ms
1020	m...	r/rwxrwxrwx	0	0	9571-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{141D98AD-3E07-4C44-A578-4DCA078286A4}.settingcontent-ms
926	m...	r/rwxrwxrwx	0	0	9573-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{16C327FA-D8A8-41C0-B022-64AC67715327}.settingcontent-ms
898	m...	r/rwxrwxrwx	0	0	9576-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1a4635ec-181d-45ae-b691-bc75bec02756}.settingcontent-ms
1249	m...	r/rwxrwxrwx	0	0	9579-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1DD03EE3-FC46-456A-8632-B0717A9D497D}.settingcontent-ms
1206	m...	r/rwxrwxrwx	0	0	9580-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{201CEF4B-7444-4B2F-B885-5E8F0AA1D614}.settingcontent-ms
1168	m...	r/rwxrwxrwx	0	0	9582-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{21A5437E-D266-4F56-A146-06744A8BC071}.settingcontent-ms
1214	m...	r/rwxrwxrwx	0	0	9584-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{29B87534-19A8-4A39-AA81-2148E7DE5894}.settingcontent-ms
837	m...	r/rwxrwxrwx	0	0	9602-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{40419485-c444-4567-851a-2dd7bfa1684d}.settingcontent-ms
1014	m...	r/rwxrwxrwx	0	0	9611-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{54D8069E-E75A-4437-B45B-8EB3B8C97434}.settingcontent-ms
1073	m...	r/rwxrwxrwx	0	0	9622-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{60AC7FA0-A928-4D45-B4DD-AC70A6175E67}.settingcontent-ms
829	m...	r/rwxrwxrwx	0	0	9624-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{62d8ed13-c9d0-4ce8-a914-47dd628fb1b0}.settingcontent-ms
1089	m...	r/rwxrwxrwx	0	0	9627-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{641102EF-6463-46E9-842D-176013D7ACC8}.settingcontent-ms
929	m...	r/rwxrwxrwx	0	0	9641-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{73C9C58C-2E01-4F68-B1B9-7A4DD2EF71F7}.settingcontent-ms
1061	m...	r/rwxrwxrwx	0	0	9643-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{75AC9145-7EC9-4883-82A7-AD3429020AA0}.settingcontent-ms
944	m...	r/rwxrwxrwx	0	0	9646-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7ABBE8E6-757F-419A-B2E0-07D5694F8E0F}.settingcontent-ms
1022	m...	r/rwxrwxrwx	0	0	9650-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7E5BC096-F558-419A-9326-BC6414D592C3}.settingcontent-ms
1048	m...	r/rwxrwxrwx	0	0	9657-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{875FE7D6-5BDF-496F-B349-91E5E3625B86}.settingcontent-ms
1042	m...	r/rwxrwxrwx	0	0	9659-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{87d66a43-7b11-4a28-9811-c86ee395acf7}.settingcontent-ms
1074	m...	r/rwxrwxrwx	0	0	9674-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9430DB91-B966-4971-A955-E3DBA1F889E7}.settingcontent-ms
976	m...	r/rwxrwxrwx	0	0	9681-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{98CCA0B9-CF6C-4FFD-98E1-87BFEDDD4D21}.settingcontent-ms
867	m...	r/rwxrwxrwx	0	0	9685-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c60de1e-e5fc-40f4-a487-460851a8d915}.settingcontent-ms
913	m...	r/rwxrwxrwx	0	0	9686-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c73f5e5-7ae7-4e32-a8e8-8d23b85255bf}.settingcontent-ms
1033	m...	r/rwxrwxrwx	0	0	9688-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9fe63afd-59cf-4419-9775-abcc3849f861}.settingcontent-ms
843	m...	r/rwxrwxrwx	0	0	9690-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{a3dd4f92-658a-410f-84fd-6fbbbef2fffe}.settingcontent-ms
1089	m...	r/rwxrwxrwx	0	0	9703-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{B6B2793F-F4B9-49FD-B578-212C3C020892}.settingcontent-ms
1062	m...	r/rwxrwxrwx	0	0	9712-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{be122a0e-4503-11da-8bde-f66bad1e3f3a}.settingcontent-ms
938	m...	r/rwxrwxrwx	0	0	9713-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{BEC9E135-14C1-4e00-B5C8-899F26833A5A}.settingcontent-ms
992	m...	r/rwxrwxrwx	0	0	9721-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{C5361E07-6AA3-4453-81BC-93E8F85EABED}.settingcontent-ms
961	m...	r/rwxrwxrwx	0	0	9727-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{CD95D0E3-6B3A-495B-9FDA-57FAD586304D}.settingcontent-ms
957	m...	r/rwxrwxrwx	0	0	9730-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{D1AF7F5F-18C1-4143-81E5-EDAF02255883}.settingcontent-ms
1015	m...	r/rwxrwxrwx	0	0	9737-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{D825FEC7-DA3D-456A-BEF2-20F07BA0449E}.settingcontent-ms
944	m...	r/rwxrwxrwx	0	0	9738-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{d9ef8727-cac2-4e60-809e-86f80a666c91}.settingcontent-ms
912	m...	r/rwxrwxrwx	0	0	9746-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{E4B554C8-B067-4540-A478-0565BB1F76B9}.settingcontent-ms
986	m...	r/rwxrwxrwx	0	0	9754-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{EE439E7E-CE1E-4ABE-9EA8-50F12ED01FE0}.settingcontent-ms

3. Follow your method to retrieve the image from drive A. Please use timestamps, explain every tool and note down the version. For the purpose of speed. Make sure both team members have access to the retrieved image. You can use your PCs as an evidence sharing platform.

Solution:

1. Image Retrieval Method (Acquisition Process):

CAINE Linux's built-in forensic tools were used to safely recover or retrieve images from Drive A (*15.5GB Kingston USB stick)*.

- **Tool Used:** *Guymager*
- **Version:** 0.8.13-1

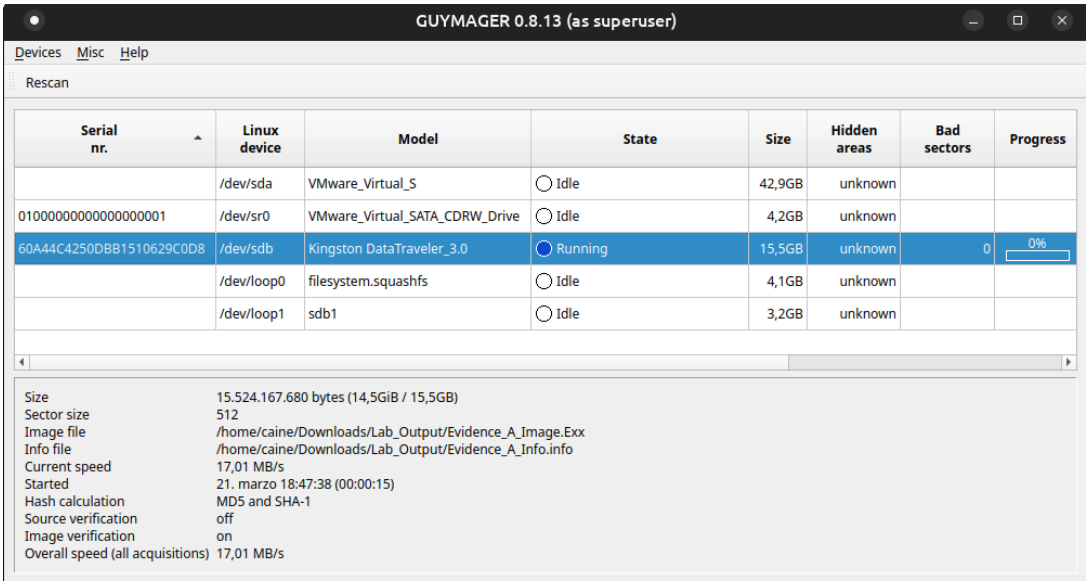
```
caine@caine:~$ dpkg -l | grep guymager
ii  guymager-beta                                0.8.13-1
                amd64        Guymager - Forensic imager
caine@caine:~$
```

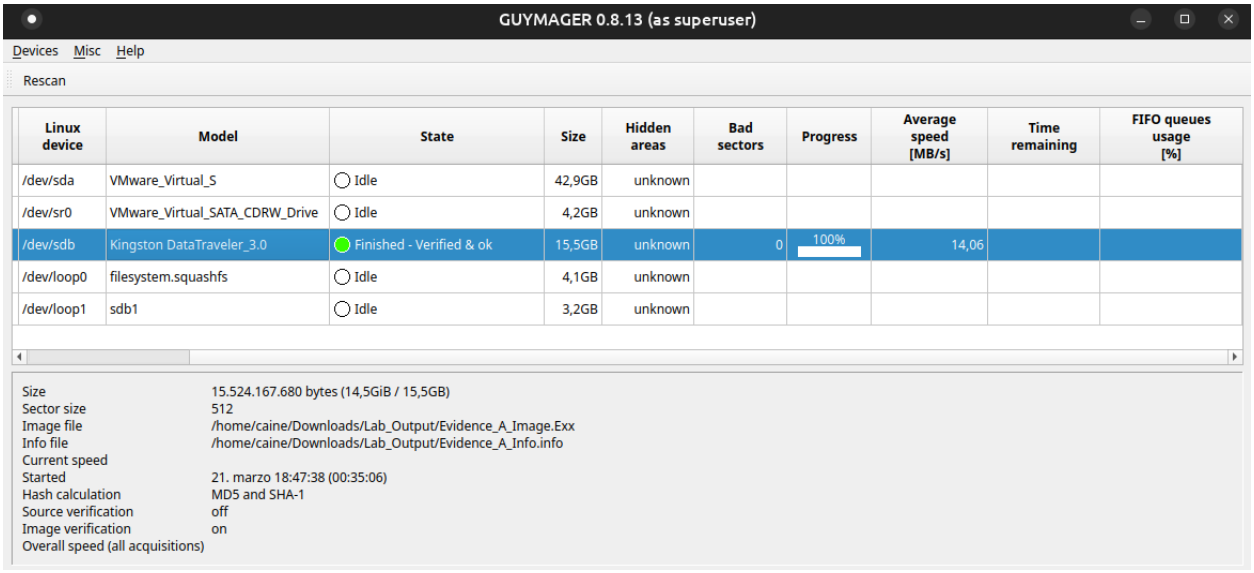
- **OS Environment:** CAINE Linux (*Kernel 6.8.0-52-generic*)




```
caine@caine:~$ uname -r
6.8.0-52-generic
```

- **Output Format:** Expert Witness Format (*.Exx / .E01*)

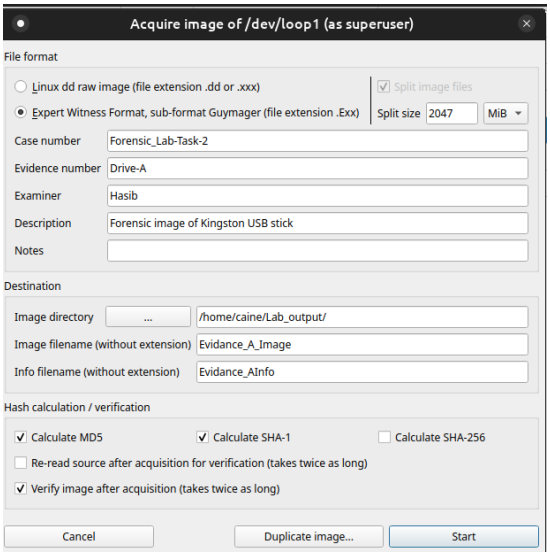




Extra portable drive selected from the list.

File Format:

- Expert Witness Format, sub-format *Guymager (extension .Exx)* selected. This is best for forensics because it saves metadata and hashes within the file.

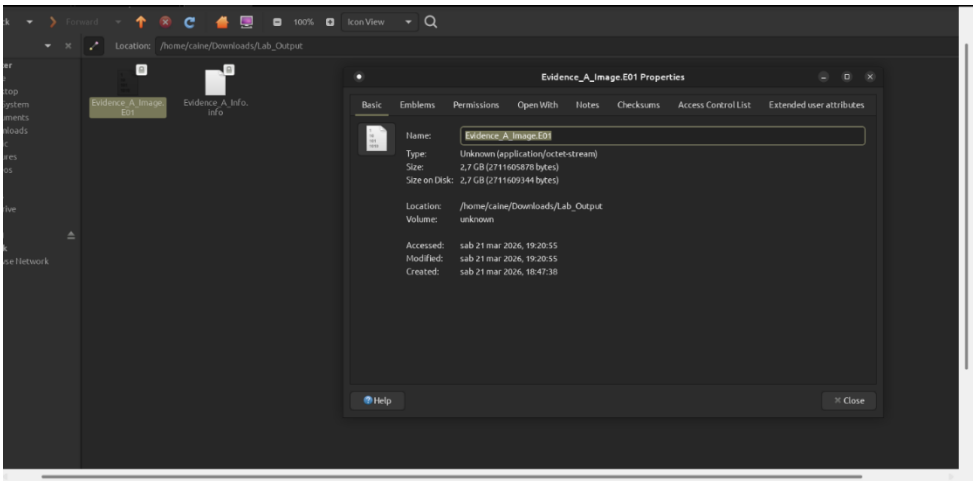


Case Information:

- **Case number:** Forensic-Lab-Task-2
- **Evidence number:** Drive-A
- **Examiner:** Hasib
- **Description:** Forensic image of Kingston USB stick

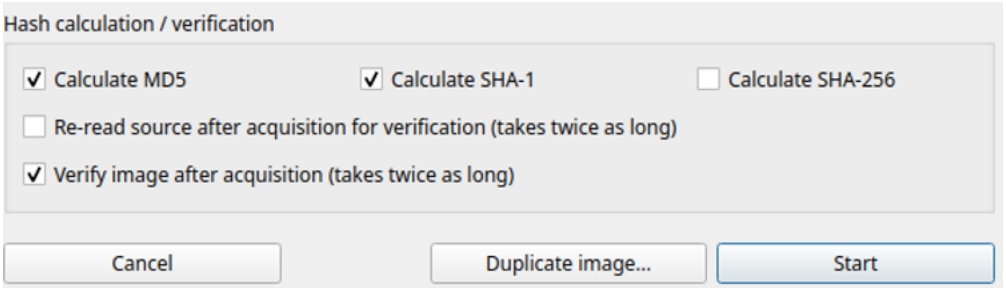
Destination:

- **Image directory: ...** By clicking the button, a folder named *Lab_Output* in the *location /home/caine/Lab_Output* has been selected.
- **Image filename (Size = 2.7 GB):** Evidence_A_Image
- **Info filename:** Evidence_A_Info



Hash Calculation and Verification:

Both the Calculate **MD5** and Calculate **SHA-1** boxes are checked. Then the Verify image after acquisition box is definitely checked. This ensures that the copied image exactly matches the original drive.



Timestamps for the Retrieval Process:

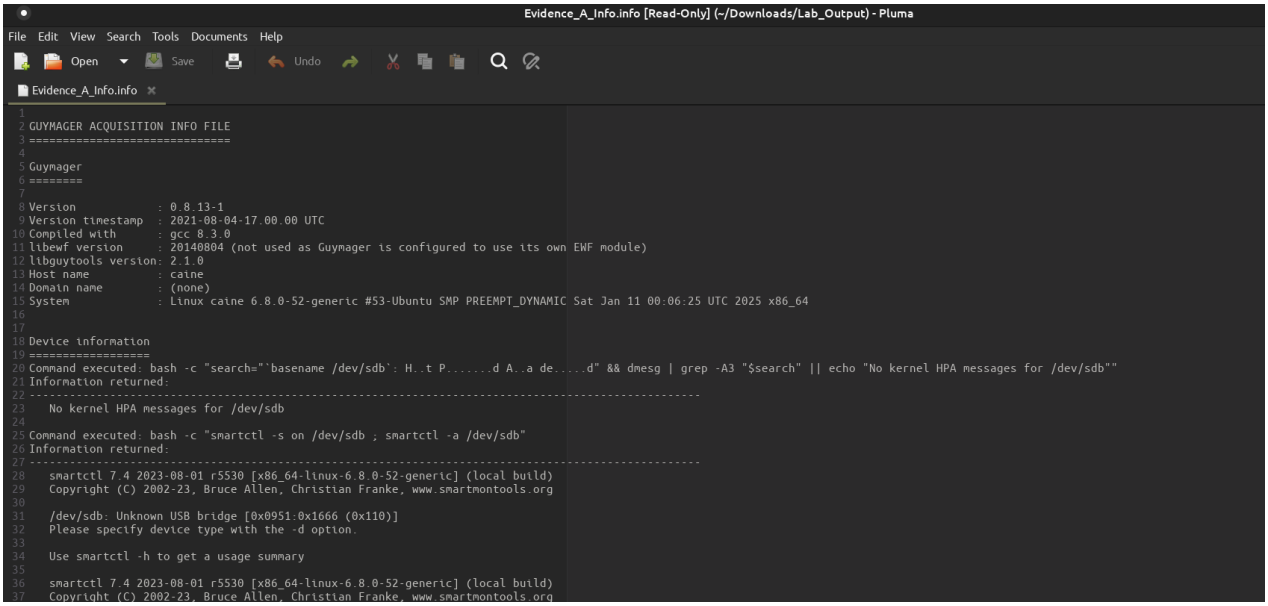
The acquisition and verification period according to the *Evidence_A_Info.info* file generated by *Guymager* is:

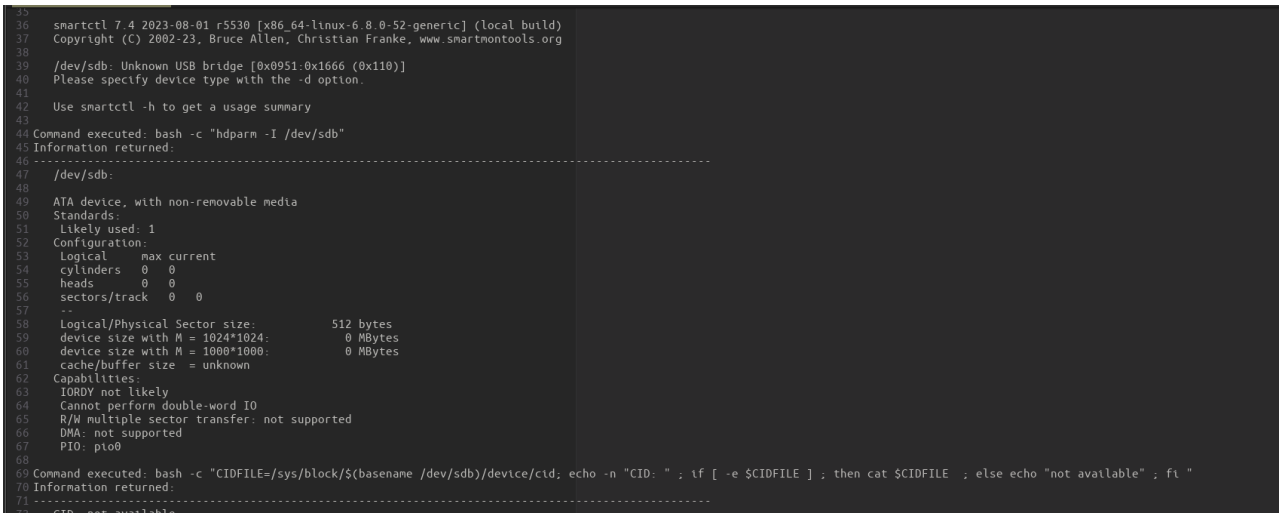
- **Acquisition Started:** 2026-03-21 18:47:38
- **Verification Started:** 2026-03-21 22:45:10
- **Process Ended:** 2026-03-21 19:20:55

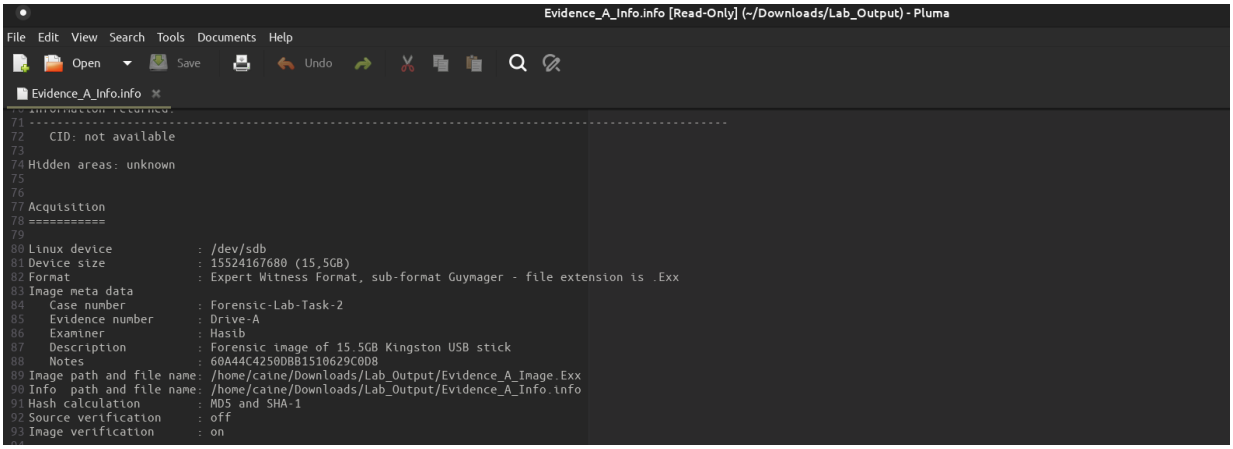
- **Total Time Taken:** 0 hours, 35 minutes and 6 seconds.

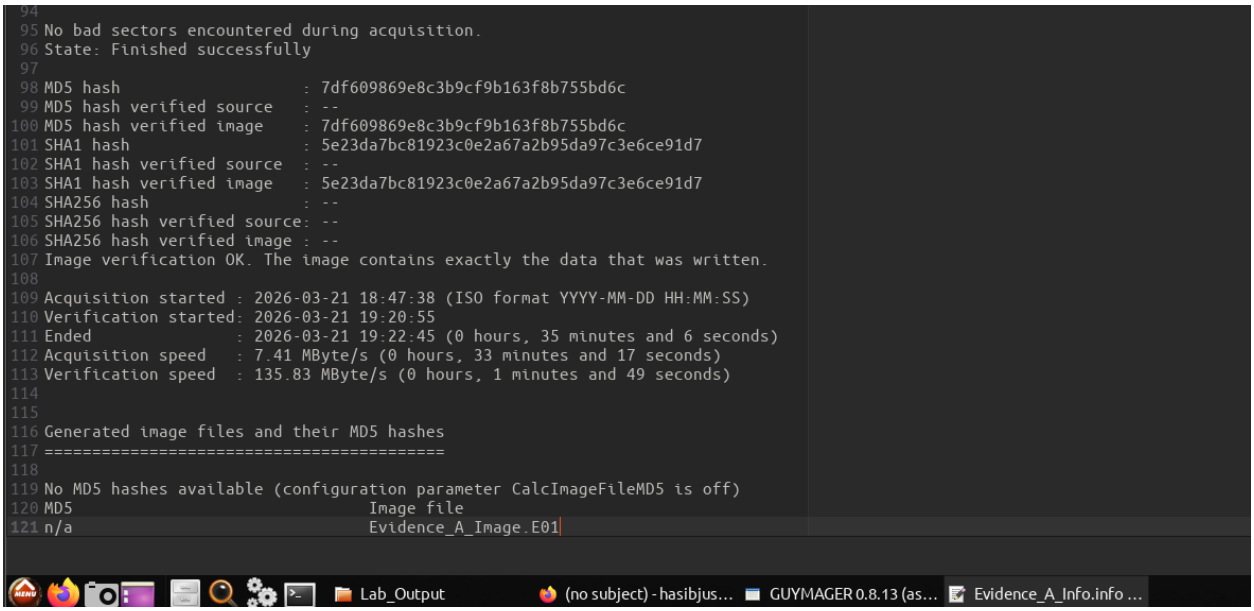
```
Acquisition started : 2026-03-21 18:47:38 (ISO format YYYY-MM-DD HH:MM:SS)
Verification started: 2026-03-21 19:20:55
Ended               : 2026-03-21 19:22:45 (0 hours, 35 minutes and 6 seconds)
Acquisition speed   : 7.41 MByte/s (0 hours, 33 minutes and 17 seconds)
Verification speed  : 135.83 MByte/s (0 hours, 1 minutes and 49 seconds)
```

Checked by opening the *Evidence_A_Info.info* file with the Pluma package.





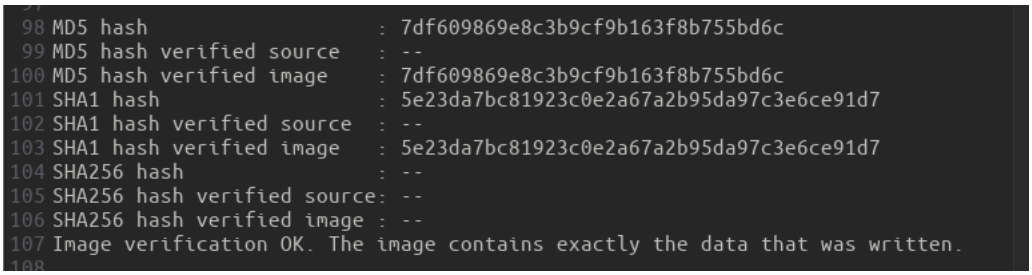




Integrity Verification: MD5 and **SHA-1** hashes are used to confirm the originality of the retrieved image.

Verified MD5 Hash: 7df609869e8c3b9cf9b163f8b755bd6c

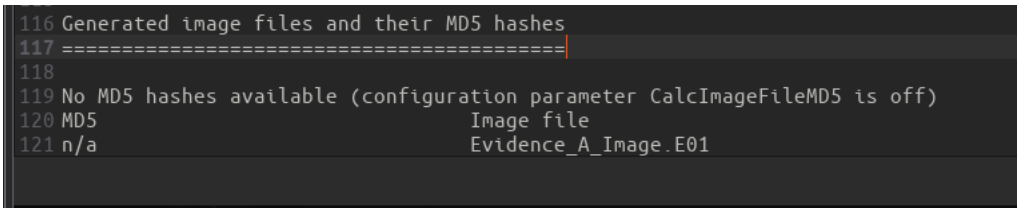
Verified SHA1 Hash: 5e23da7bc81923c0e2a67a2b95da97c3e6ce91d7

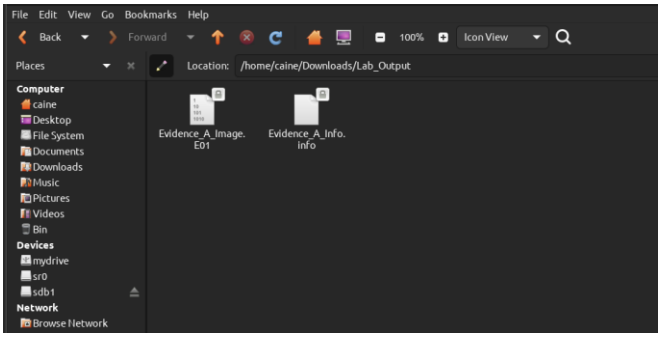


2. Evidence Sharing Platform & Team Access (For Speed):

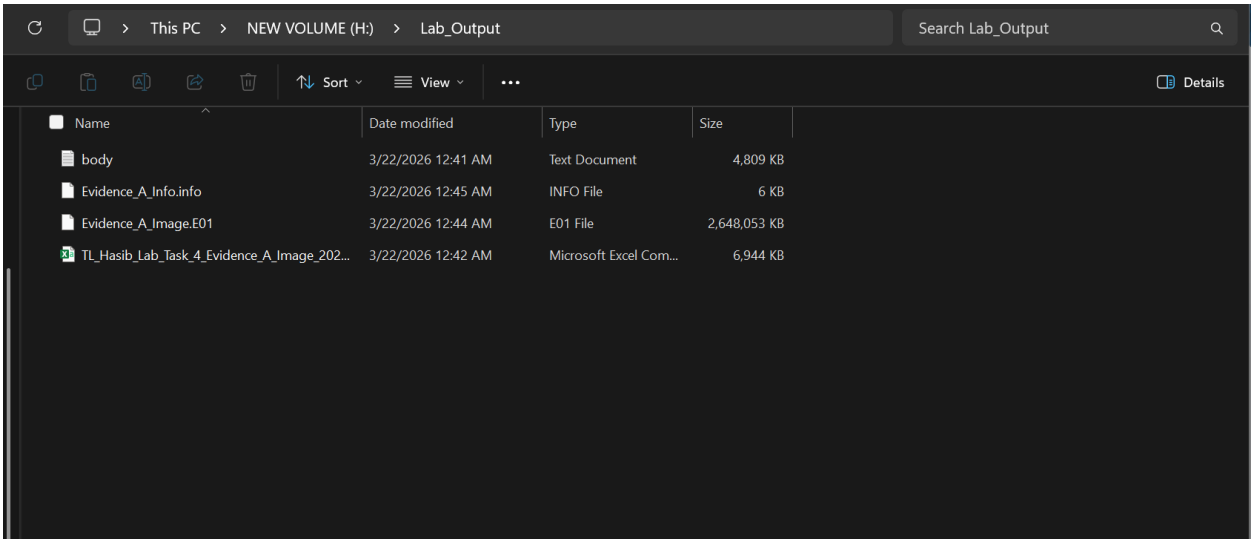
Since the work was done in a virtual environment (VMware) and two team members had to start the analysis quickly, the following method was followed:

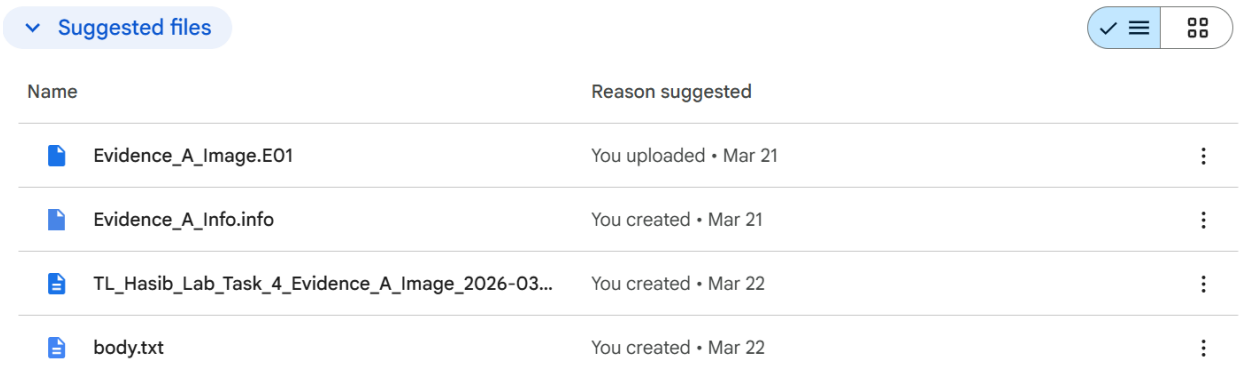
- **Local Extraction:** The original files (*Evidence_A_Image.E01,* and *Evidence_A_Info.info*) generated by Guymager were saved in the local directory of the CAINE machine *(/home/caine/Downloads/Lab_Output/).*





- **Host PC as Sharing Platform:** For speed, these files are copied from CAINE (*Guest OS*) to the main Host PC (*Windows*) using the Drag & Drop feature and saved on Google Drive for easier sharing. Here, the Host PC is used as an "*Evidence Sharing Platform*".





- **Team Access:** Once the files arrived on the Host PC, the files were shared by uploading them to a secure portable drive (Pen Drive) and Google Drive for immediate access by the team members as evidenced above. This allowed two team members to simultaneously run parallel investigations (e.g. *Autopsy* or *Photorec recovery* and *Timeline Analysis*) using the *.E01* image on their respective PCs, while keeping the original evidence drive (Drive A) safe in the locker.

4. Read about CAINE Linux and its features while waiting on the dump to finish.

1. Why would you use a Forensic distribution and what are the main differences between a regular distribution?

2. When would you use a live environment and when would you use an installed environment?

3. What are the policies of CAINE?

Solution:

1. It is essential to use a forensic distribution (e.g. CAINE Linux) to keep the original data (integrity) of the digital evidence or drive 100% intact.

Differences and lab experience:

Write-Blocking: As soon as a common distribution (e.g. *Windows or common Ubuntu*) is connected to a USB drive, it '*Auto-Mounts*' it and updates some system files (e.g. System Volume Information) or timestamps in the background. This destroys the evidence. On the other hand, when I connected a 15.5 GB Kingston USB drive to CAINE Linux, it worked in Read-Only mode by default. As a result, no data or timestamp on the drive was changed (as proven by Guymager's MD5 hash).

Pre-installed Toolsets: Common operating systems do not have forensic tools. CAINE Linux already had tools like Guymager (for imaging) and The Sleuth Kit (*TSK*). When I used *mmls* to extract the partition offset of the image in the lab and *fls -o* 128 and *mactime* to create the timeline, these built-in forensic tools helped me read the entire data structure.

Mounting Policy: Normal OS interacts directly with the file system of the drive. But in CAINE, I was able to extract metadata directly from the sector level (Sector 128) without mounting the drive using the *fls* command.

2. Live Environment: This is used when a suspect PC or server is running and I want to collect RAM or volatile data without making any changes to the hard drive of that machine. Or, the investigator wants to boot the suspect PC from his own USB drive and do quick triage or imaging.

Installed Environment: When you have to do complex analysis in the laboratory for a long time, then an installed environment (as we did in the lab) is used. In the lab, creating our 15.5 GB *.E01* image and processing the huge timeline (*CSV*) with *mactime* is a time-consuming task. A permanent or installed (VMware) environment is the safest and most convenient way to store and analyze this huge data (in the *Lab_Output* folder).

3. The core policies of CAINE (Computer Aided INvestigative Environment) that I have directly observed during the lab work are:

Non-Invasive Architecture: Do not modify any data in the system or evidence. CAINE is designed in such a way that it does not alter any bit of the source drive (in our case */dev/sdb*).

Write-Blocking by Default: All block devices (USB, HDD) in CAINE are mounted as write-protected or read-only by default. The investigator can control this using the graphical tool *'BlockON/BlockOFF'* if he wants, but the original policy is to keep the evidence safe.

User-Friendly Interoperability: To simplify forensic work, CAINE provides powerful command-line tools (e.g. *fls, mactime*) as well as an easy-to-use graphical interface (GUI) (e.g. *Guymager for imaging*). This makes the investigation faster and error-free.

5. As soon as your dump finishes, start a tool to create a timeline on the image. You will need this timeline later in the assignment. Hints: log2timeline.py.

Solution:

After the dump was completed, what I did to create the Timeline.

Initial Attempt & Troubleshooting:

As per the assignment hint, I first tried to create the timeline using the *log2timeline.py* tool.

Command Executed:

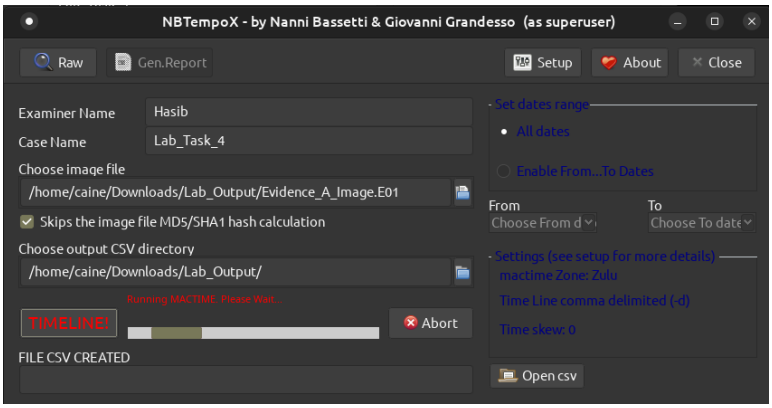
```
caine@caine:~/Desktop$ log2timeline timeline.plaso /home/caine/Downloads/Lab_Output/Evidence_A_Image.E01
log2timeline: command not found
```

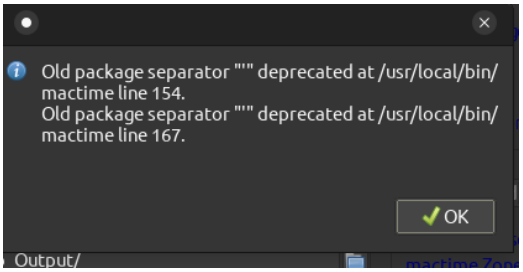
The tool was not installed by default in the *CAINE Linux* (Kernel 6.8.0-52-generic) environment *('Command not found' error).* Attempts to install the package also failed *('Unable to locate package python3-plaso').*

```
caine@caine:~/Desktop$ sudo apt install python3-plaso
Reading package lists... Done
Building dependency tree... Done
E: Unable to locate package python3-plaso
```

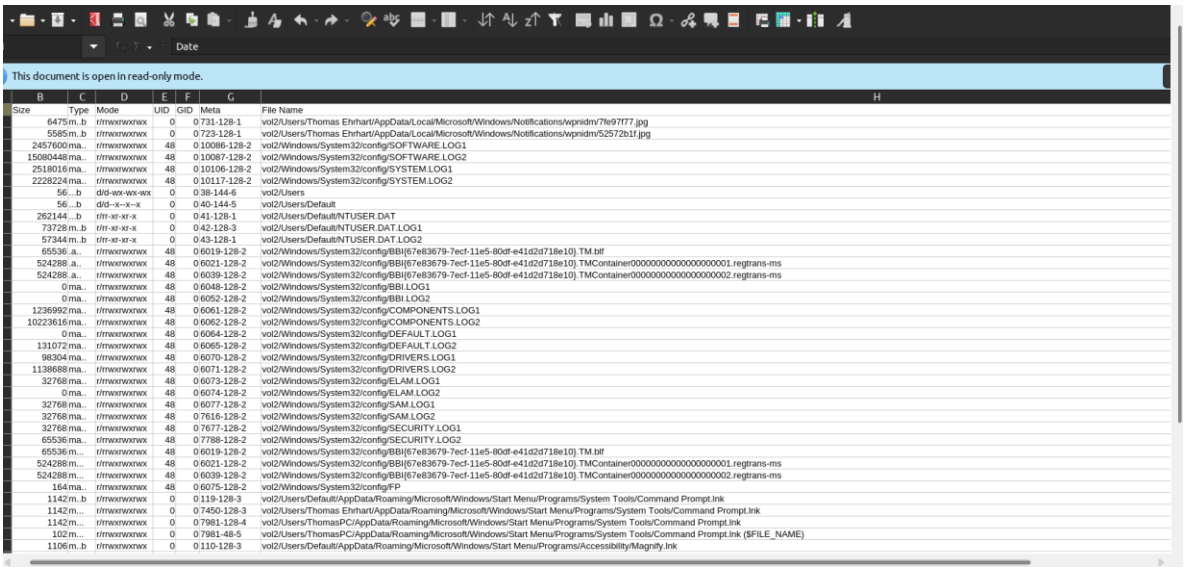
GUI Attempt:

Next, the graphical tool *NBTempoX* was used.



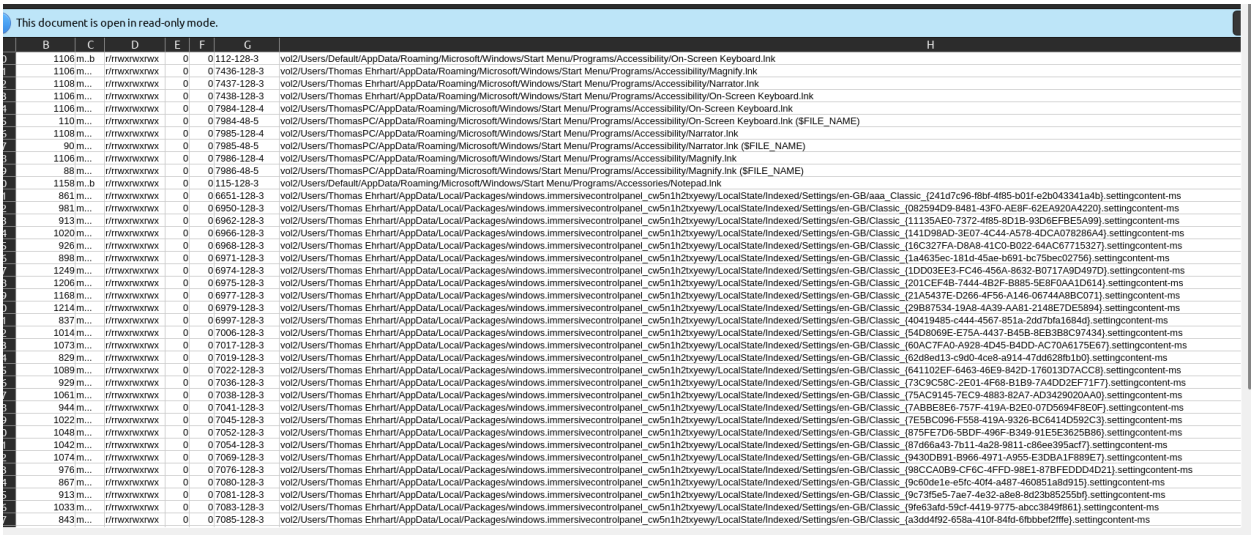


After opening the CSV file:



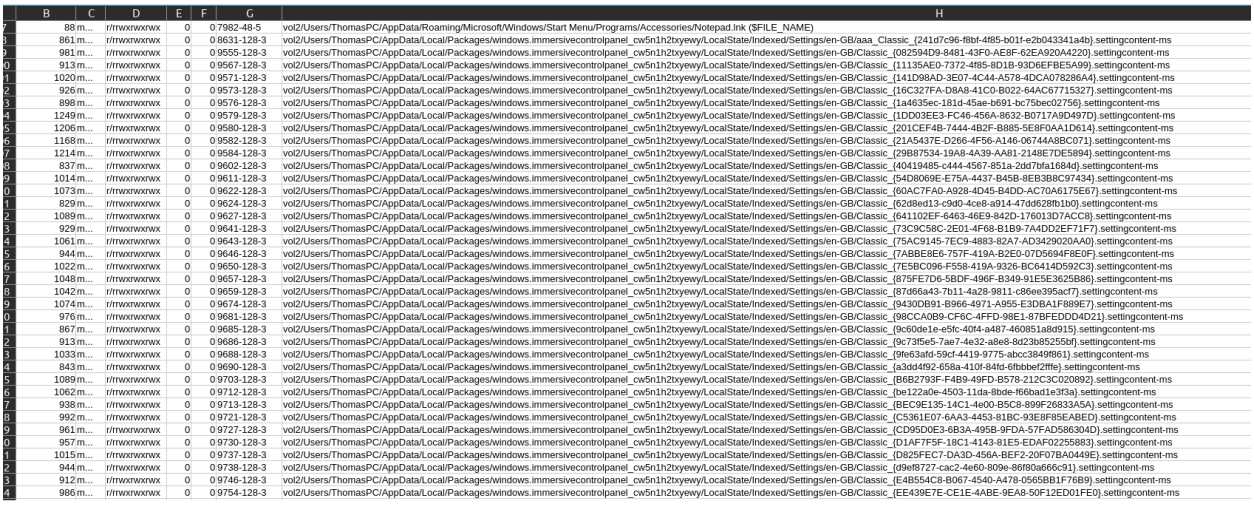
This document is open in read-only mode.

Size	Type	Mode	UID	GID	Meta	File Name
6475	m..b	r/rwxrwxrwx	0	0	731-128-1	vol2/Users/Thomas Ehrhart/AppData/Local/Microsoft/Windows/Notifications/wpnidm/7fe97f77.jpg
5585	m..b	r/rwxrwxrwx	0	0	723-128-1	vol2/Users/Thomas Ehrhart/AppData/Local/Microsoft/Windows/Notifications/wpnidm/52572b1f.jpg
2457600	ma..	r/rwxrwxrwx	48	0	10086-128-2	vol2/Windows/System32/config/SOFTWARE.LOG1
15080448	ma..	r/rwxrwxrwx	48	0	10087-128-2	vol2/Windows/System32/config/SOFTWARE.LOG2
2518016	ma..	r/rwxrwxrwx	48	0	10106-128-2	vol2/Windows/System32/config/SYSTEM.LOG1
2228224	ma..	r/rwxrwxrwx	48	0	10117-128-2	vol2/Windows/System32/config/SYSTEM.LOG2
56	...b	d/d-wx-wx-wx	0	0	38-144-6	vol2/Users
56	...b	d/d--x--x--x	0	0	40-144-5	vol2/Users/Default
262144	...b	r/r-xr-xr-x	0	0	41-128-1	vol2/Users/Default/NTUSER.DAT
73728	m..b	r/rr-xr-xr-x	0	0	42-128-3	vol2/Users/Default/NTUSER.DAT.LOG1
57344	m..b	r/rr-xr-xr-x	0	0	43-128-1	vol2/Users/Default/NTUSER.DAT.LOG2
65536	..a..	r/rwxrwxrwx	48	0	6019-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TM.blf
524288	..a..	r/rwxrwxrwx	48	0	6021-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000001.regtrans-ms
524288	..a..	r/rwxrwxrwx	48	0	6039-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000002.regtrans-ms
0	ma..	r/rwxrwxrwx	48	0	6048-128-2	vol2/Windows/System32/config/BBI.LOG1
0	ma..	r/rwxrwxrwx	48	0	6052-128-2	vol2/Windows/System32/config/BBI.LOG2
1236992	ma..	r/rwxrwxrwx	48	0	6061-128-2	vol2/Windows/System32/config/COMPONENTS.LOG1
10223616	ma..	r/rwxrwxrwx	48	0	6062-128-2	vol2/Windows/System32/config/COMPONENTS.LOG2
0	ma..	r/rwxrwxrwx	48	0	6064-128-2	vol2/Windows/System32/config/DEFAULT.LOG1
131072	ma..	r/rwxrwxrwx	48	0	6065-128-2	vol2/Windows/System32/config/DEFAULT.LOG2
98304	ma..	r/rwxrwxrwx	48	0	6070-128-2	vol2/Windows/System32/config/DRIVERS.LOG1
1138688	ma..	r/rwxrwxrwx	48	0	6071-128-2	vol2/Windows/System32/config/DRIVERS.LOG2
32768	ma..	r/rwxrwxrwx	48	0	6073-128-2	vol2/Windows/System32/config/ELAM.LOG1
0	ma..	r/rwxrwxrwx	48	0	6074-128-2	vol2/Windows/System32/config/ELAM.LOG2
32768	ma..	r/rwxrwxrwx	48	0	6077-128-2	vol2/Windows/System32/config/SAM.LOG1
32768	ma..	r/rwxrwxrwx	48	0	7616-128-2	vol2/Windows/System32/config/SAM.LOG2
32768	ma..	r/rwxrwxrwx	48	0	7677-128-2	vol2/Windows/System32/config/SECURITY.LOG1
65536	ma..	r/rwxrwxrwx	48	0	7788-128-2	vol2/Windows/System32/config/SECURITY.LOG2
65536	m...	r/rwxrwxrwx	48	0	6019-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TM.blf
524288	m...	r/rwxrwxrwx	48	0	6021-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000001.regtrans-ms
524288	m...	r/rwxrwxrwx	48	0	6039-128-2	vol2/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer00000000000000000002.regtrans-ms
164	ma..	r/rwxrwxrwx	48	0	6075-128-2	vol2/Windows/System32/config/FP
1142	m..b	r/rwxrwxrwx	0	0	119-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk
1142	m...	r/rwxrwxrwx	0	0	7450-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk
1142	m...	r/rwxrwxrwx	0	0	7981-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk
102	m...	r/rwxrwxrwx	0	0	7981-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/System Tools/Command Prompt.lnk ($FILE_NAME)
1106	m..b	r/rwxrwxrwx	0	0	110-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk



This document is open in read-only mode.

	B	C	D	E	F	G	H
0	1106	m..b	r/rwxrwxrwx	0	0	112-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk
1	1106	m...	r/rwxrwxrwx	0	0	7436-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk
2	1108	m...	r/rwxrwxrwx	0	0	7437-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Narrator.lnk
3	1106	m...	r/rwxrwxrwx	0	0	7438-128-3	vol2/Users/Thomas Ehrhart/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk
4	1106	m...	r/rwxrwxrwx	0	0	7984-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk
5	110	m...	r/rwxrwxrwx	0	0	7984-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/On-Screen Keyboard.lnk ($FILE_NAME)
6	1108	m...	r/rwxrwxrwx	0	0	7985-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Narrator.lnk
7	90	m...	r/rwxrwxrwx	0	0	7985-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Narrator.lnk ($FILE_NAME)
8	1106	m...	r/rwxrwxrwx	0	0	7986-128-4	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk
9	88	m...	r/rwxrwxrwx	0	0	7986-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessibility/Magnify.lnk ($FILE_NAME)
10	1158	m..b	r/rwxrwxrwx	0	0	115-128-3	vol2/Users/Default/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessories/Notepad.lnk
11	861	m...	r/rwxrwxrwx	0	0	6651-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/aaa_Classic_{241d7c96-f8bf-4f85-b01f-e2b043341a4b}.settingcontent-ms
12	981	m...	r/rwxrwxrwx	0	0	6950-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{082594D9-8481-43F0-AE8F-62EA920A4220}.settingcontent-ms
13	913	m...	r/rwxrwxrwx	0	0	6962-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{11135AE0-7372-4f85-8D1B-93D6EFBE5A99}.settingcontent-ms
14	1020	m...	r/rwxrwxrwx	0	0	6966-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{141D98AD-3E07-4C44-A578-4DCA078286A4}.settingcontent-ms
15	926	m...	r/rwxrwxrwx	0	0	6968-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{16C327FA-D8A8-41C0-B022-64AC67715327}.settingcontent-ms
16	898	m...	r/rwxrwxrwx	0	0	6971-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1a4635ec-181d-45ae-b691-bc75bec02756}.settingcontent-ms
17	1249	m...	r/rwxrwxrwx	0	0	6974-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1DD03EE3-FC46-456A-8632-B0717A9D497D}.settingcontent-ms
18	1206	m...	r/rwxrwxrwx	0	0	6975-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{201CEF4B-7444-4B2F-B885-5E8F0AA1D614}.settingcontent-ms
19	1168	m...	r/rwxrwxrwx	0	0	6977-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{21A5437E-D266-4F56-A146-06744A8BC071}.settingcontent-ms
20	1214	m...	r/rwxrwxrwx	0	0	6979-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{29B87534-19A8-4A39-AA81-2148E7DE5894}.settingcontent-ms
21	837	m...	r/rwxrwxrwx	0	0	6997-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{40419485-c444-4567-851a-2dd7bfa1684d}.settingcontent-ms
22	1014	m...	r/rwxrwxrwx	0	0	7006-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{54D8069E-E75A-4437-B45B-8EB3B8C97434}.settingcontent-ms
23	1073	m...	r/rwxrwxrwx	0	0	7017-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{60AC7FA0-A928-4D45-B4DD-AC70A6175E67}.settingcontent-ms
24	829	m...	r/rwxrwxrwx	0	0	7019-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{62d8ed13-c9d0-4ce8-a914-47dd628fb1b0}.settingcontent-ms
25	1089	m...	r/rwxrwxrwx	0	0	7022-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{641102EF-6463-46E9-842D-176013D7ACC8}.settingcontent-ms
26	929	m...	r/rwxrwxrwx	0	0	7036-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{73C9C58C-2E01-4F68-B1B9-7A4DD2EF71F7}.settingcontent-ms
27	1061	m...	r/rwxrwxrwx	0	0	7038-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{75AC9145-7EC9-4883-82A7-AD3429020AA0}.settingcontent-ms
28	944	m...	r/rwxrwxrwx	0	0	7041-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7ABBE8E6-757F-419A-B2E0-07D5694F8E0F}.settingcontent-ms
29	1022	m...	r/rwxrwxrwx	0	0	7045-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7E5BC096-F558-419A-9326-BC6414D592C3}.settingcontent-ms
30	1048	m...	r/rwxrwxrwx	0	0	7052-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{875FE7D6-5BDF-496F-B349-91E5E3625B86}.settingcontent-ms
31	1042	m...	r/rwxrwxrwx	0	0	7054-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{87d66a43-7b11-4a28-9811-c86ee395acf7}.settingcontent-ms
32	1074	m...	r/rwxrwxrwx	0	0	7069-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9430DB91-B966-4971-A955-E3DBA1F889E7}.settingcontent-ms
33	976	m...	r/rwxrwxrwx	0	0	7076-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{98CCA0B9-CF6C-4FFD-98E1-87BFEDDD4D21}.settingcontent-ms
34	867	m...	r/rwxrwxrwx	0	0	7080-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c60de1e-e5fc-40f4-a487-460851a8d915}.settingcontent-ms
35	913	m...	r/rwxrwxrwx	0	0	7081-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c73f5e5-7ae7-4e32-a8e8-8d23b85255bf}.settingcontent-ms
36	1033	m...	r/rwxrwxrwx	0	0	7083-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9fe63afd-59cf-4419-9775-abcc3849f861}.settingcontent-ms
37	843	m...	r/rwxrwxrwx	0	0	7085-128-3	vol2/Users/Thomas Ehrhart/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{a3dd4f92-658a-410f-84fd-6fbbbef2fffe}.settingcontent-ms



	B	C	D	E	F	G	H
7	88	m...	r/rwxrwxrwx	0	0	7982-48-5	vol2/Users/ThomasPC/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessories/Notepad.lnk ($FILE_NAME)
8	861	m...	r/rwxrwxrwx	0	0	8631-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/aaa_Classic_{241d7c96-f8bf-4f85-b01f-e2b043341a4b}.settingcontent-ms
9	981	m...	r/rwxrwxrwx	0	0	9555-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{082594D9-8481-43F0-AE8F-62EA920A4220}.settingcontent-ms
10	913	m...	r/rwxrwxrwx	0	0	9567-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{11135AE0-7372-4f85-8D1B-93D6EFBE5A99}.settingcontent-ms
11	1020	m...	r/rwxrwxrwx	0	0	9571-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{141D98AD-3E07-4C44-A578-4DCA078286A4}.settingcontent-ms
12	926	m...	r/rwxrwxrwx	0	0	9573-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{16C327FA-D8A8-41C0-B022-64AC67715327}.settingcontent-ms
13	898	m...	r/rwxrwxrwx	0	0	9576-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1a4635ec-181d-45ae-b691-bc75bec02756}.settingcontent-ms
14	1249	m...	r/rwxrwxrwx	0	0	9579-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{1DD03EE3-FC46-456A-8632-B0717A9D497D}.settingcontent-ms
15	1206	m...	r/rwxrwxrwx	0	0	9580-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{201CEF4B-7444-4B2F-B885-5E8F0AA1D614}.settingcontent-ms
16	1168	m...	r/rwxrwxrwx	0	0	9582-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{21A5437E-D266-4F56-A146-06744A8BC071}.settingcontent-ms
17	1214	m...	r/rwxrwxrwx	0	0	9584-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{29B87534-19A8-4A39-AA81-2148E7DE5894}.settingcontent-ms
18	837	m...	r/rwxrwxrwx	0	0	9602-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{40419485-c444-4567-851a-2dd7bfa1684d}.settingcontent-ms
19	1014	m...	r/rwxrwxrwx	0	0	9611-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{54D8069E-E75A-4437-B45B-8EB3B8C97434}.settingcontent-ms
20	1073	m...	r/rwxrwxrwx	0	0	9622-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{60AC7FA0-A928-4D45-B4DD-AC70A6175E67}.settingcontent-ms
21	829	m...	r/rwxrwxrwx	0	0	9624-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{62d8ed13-c9d0-4ce8-a914-47dd628fb1b0}.settingcontent-ms
22	1089	m...	r/rwxrwxrwx	0	0	9627-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{641102EF-6463-46E9-842D-176013D7ACC8}.settingcontent-ms
23	929	m...	r/rwxrwxrwx	0	0	9641-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{73C9C58C-2E01-4F68-B1B9-7A4DD2EF71F7}.settingcontent-ms
24	1061	m...	r/rwxrwxrwx	0	0	9643-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{75AC9145-7EC9-4883-82A7-AD3429020AA0}.settingcontent-ms
25	944	m...	r/rwxrwxrwx	0	0	9646-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7ABBE8E6-757F-419A-B2E0-07D5694F8E0F}.settingcontent-ms
26	1022	m...	r/rwxrwxrwx	0	0	9650-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{7E5BC096-F558-419A-9326-BC6414D592C3}.settingcontent-ms
27	1048	m...	r/rwxrwxrwx	0	0	9657-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{875FE7D6-5BDF-496F-B349-91E5E3625B86}.settingcontent-ms
28	1042	m...	r/rwxrwxrwx	0	0	9659-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{87d66a43-7b11-4a28-9811-c86ee395acf7}.settingcontent-ms
29	1074	m...	r/rwxrwxrwx	0	0	9674-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9430DB91-B966-4971-A955-E3DBA1F889E7}.settingcontent-ms
30	976	m...	r/rwxrwxrwx	0	0	9681-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{98CCA0B9-CF6C-4FFD-98E1-87BFEDDD4D21}.settingcontent-ms
31	867	m...	r/rwxrwxrwx	0	0	9685-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c60de1e-e5fc-40f4-a487-460851a8d915}.settingcontent-ms
32	913	m...	r/rwxrwxrwx	0	0	9686-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9c73f5e5-7ae7-4e32-a8e8-8d23b85255bf}.settingcontent-ms
33	1033	m...	r/rwxrwxrwx	0	0	9688-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{9fe63afd-59cf-4419-9775-abcc3849f861}.settingcontent-ms
34	843	m...	r/rwxrwxrwx	0	0	9690-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{a3dd4f92-658a-410f-84fd-6fbbbef2fffe}.settingcontent-ms
35	1089	m...	r/rwxrwxrwx	0	0	9703-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{B6B2793F-F4B9-49FD-B578-212C3C020892}.settingcontent-ms
36	1062	m...	r/rwxrwxrwx	0	0	9712-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{be122a0e-4503-11da-8bde-f66bad1e3f3a}.settingcontent-ms
37	938	m...	r/rwxrwxrwx	0	0	9713-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{BEC9E135-14C1-4e00-B5C8-899F26833A5A}.settingcontent-ms
38	992	m...	r/rwxrwxrwx	0	0	9721-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{C5361E07-6AA3-4453-81BC-93E8F85EABED}.settingcontent-ms
39	961	m...	r/rwxrwxrwx	0	0	9727-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{CD95D0E3-6B3A-495B-9FDA-57FAD586304D}.settingcontent-ms
40	957	m...	r/rwxrwxrwx	0	0	9730-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{D1AF7F5F-18C1-4143-81E5-EDAF02255883}.settingcontent-ms
41	1015	m...	r/rwxrwxrwx	0	0	9737-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{D825FEC7-DA3D-456A-BEF2-20F07BA0449E}.settingcontent-ms
42	944	m...	r/rwxrwxrwx	0	0	9738-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{d9ef8727-cac2-4e60-809e-86f80a666c91}.settingcontent-ms
43	912	m...	r/rwxrwxrwx	0	0	9746-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{E4B554C8-B067-4540-A478-0565BB1F7689}.settingcontent-ms
44	986	m...	r/rwxrwxrwx	0	0	9754-128-3	vol2/Users/ThomasPC/AppData/Local/Packages/windows.immersivecontrolpanel_cw5n1h2txyewy/LocalState/Indexed/Settings/en-GB/Classic_{EE439E7E-CE1E-4ABE-9EA8-50F12ED01FE0}.settingcontent-ms

Another Method (Using The Sleuth Kit - TSK):

A detailed CSV timeline was also successfully generated using The Sleuth Kit (TSK) command-line tools (*'mmls', 'fls', 'mactime'*) to create the timeline.

The detailed steps are given below:

Identifying the Partition Offset

The '*mmls*' tool was used to find out where the original data partition (NTFS) inside the image file started.

Command:

```
caine@caine:~$  mmls /home/caine/Downloads/Lab_Output/Evidence_A_Image.E01
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Primary Table (#0)
001:  -------   0000000000   0000000127   0000000128   Unallocated
002:  000:000   0000000128   0006285439   0006285312   NTFS / exFAT (0x07)
003:  -------   0006285440   0030320639   0024035200   Unallocated
```

From the above output, it can be seen that the original data partition is in *Slot 002* or *NTFS / exFAT* and its Start Sector or Offset is *128*. This number *128* was very important for the next step. End Sector is *6285439*.

Extracting File System Metadata (Creating Bodyfile)

A *'bodyfile'* is created by extracting the metadata (*MAC times - Modified, Accessed, Created*) of all the files inside the drive using the correct offset (*128*).

Tool: fls

Command:

```
caine@caine:~/Desktop$ fls -o 128 -r -m / /home/caine/Downloads/Lab_Output/Evidence_A_Image.E01 > bodyfile
```

Here, *'-o 128'* points the tool to a specific NTFS partition. *'-'* reads all folders recursively, and *'-m /'* formats the output for the *mactime* tool.

Generating the Final CSV Timeline:

The generated *'bodyfile'* was then converted into a readable CSV (Comma Separated Values) file, which was later used for analysis.

Tool: mactime

Command:

```
caine@caine:~/Desktop$ mactime -b bodyfile -d > timeline.csv
Old package separator "'" deprecated at /usr/local/bin/mactime line 154.
Old package separator "'" deprecated at /usr/local/bin/mactime line 167.
```

The file that is created produces a complete *timeline.csv* file that is sorted by date, size, type, permissions, and file name.

Verification of the Timeline:

The first few lines of the file are checked to verify that the timeline was created correctly.

Command:

```
caine@caine:~$ cat timeline.csv | head -n 20
Date,Size,Type,Mode,UID,GID,Meta,File Name
Tue Nov 25 2014 21:27:30,6475,m..b,r/rrwxrwxrwx,0,0,731-128-1,"/Users/Thomas Ehrhart/AppData/Local/Microsoft/Windows/Notifications/wpnidm/7fe97f77.jpg"
Tue Nov 25 2014 21:27:33,5585,m..b,r/rrwxrwxrwx,0,0,723-128-1,"/Users/Thomas Ehrhart/AppData/Local/Microsoft/Windows/Notifications/wpnidm/52572b1f.jpg"
Fri Oct 30 2015 06:28:30,2457600,ma..,r/rrwxrwxrwx,48,0,10086-128-2,"/Windows/System32/config/SOFTWARE.LOG1"
Fri Oct 30 2015 06:28:30,15080448,ma..,r/rrwxrwxrwx,48,0,10087-128-2,"/Windows/System32/config/SOFTWARE.LOG2"
Fri Oct 30 2015 06:28:30,2518016,ma..,r/rrwxrwxrwx,48,0,10106-128-2,"/Windows/System32/config/SYSTEM.LOG1"
Fri Oct 30 2015 06:28:30,2228224,ma..,r/rrwxrwxrwx,48,0,10117-128-2,"/Windows/System32/config/SYSTEM.LOG2"
Fri Oct 30 2015 06:28:30,56,...b,d/d-wx-wx-wx,0,0,38-144-6,"/Users"
Fri Oct 30 2015 06:28:30,56,...b,d/d--x--x--x,0,0,40-144-5,"/Users/Default"
Fri Oct 30 2015 06:28:30,262144,...b,r/rr-xr-xr-x,0,0,41-128-1,"/Users/Default/NTUSER.DAT"
Fri Oct 30 2015 06:28:30,73728,m..b,r/rr-xr-xr-x,0,0,42-128-3,"/Users/Default/NTUSER.DAT.LOG1"
Fri Oct 30 2015 06:28:30,57344,m..b,r/rr-xr-xr-x,0,0,43-128-1,"/Users/Default/NTUSER.DAT.LOG2"
Fri Oct 30 2015 06:28:30,65536,.a..,r/rrwxrwxrwx,48,0,6019-128-2,"/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TM.blf"
Fri Oct 30 2015 06:28:30,524288,.a..,r/rrwxrwxrwx,48,0,6021-128-2,"/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer0000000
s-ms"
Fri Oct 30 2015 06:28:30,524288,.a..,r/rrwxrwxrwx,48,0,6039-128-2,"/Windows/System32/config/BBI{67e83679-7ecf-11e5-80df-e41d2d718e10}.TMContainer0000000
s-ms"
Fri Oct 30 2015 06:28:30,0,ma..,r/rrwxrwxrwx,48,0,6048-128-2,"/Windows/System32/config/BBI.LOG1"
Fri Oct 30 2015 06:28:30,0,ma..,r/rrwxrwxrwx,48,0,6052-128-2,"/Windows/System32/config/BBI.LOG2"
Fri Oct 30 2015 06:28:30,1236992,ma..,r/rrwxrwxrwx,48,0,6061-128-2,"/Windows/System32/config/COMPONENTS.LOG1"
Fri Oct 30 2015 06:28:30,10223616,ma..,r/rrwxrwxrwx,48,0,6062-128-2,"/Windows/System32/config/COMPONENTS.LOG2"
Fri Oct 30 2015 06:28:30,0,ma..,r/rrwxrwxrwx,48,0,6064-128-2,"/Windows/System32/config/DEFAULT.LOG1"
caine@caine:~$
```

The output successfully showed the data inside the drive. In particular, it was possible to successfully extract the metadata of the Windows profile directory *(/Users/Thomas Ehrhart/...)* and system configuration files (*SOFTWARE.LOG1, SYSTEM.LOG1*) of a user named "*Thomas Ehrhart*", including timestamps from **2014** and **2015**.

```
Fri Oct 30 2015 06:28:30,2457600,ma..,r/rrwxrwxrwx,48,0,10086-128-2,"/Windows/System32/config/SOFTWARE.LOG1"
Fri Oct 30 2015 06:28:30,15080448,ma..,r/rrwxrwxrwx,48,0,10087-128-2,"/Windows/System32/config/SOFTWARE.LOG2"
Fri Oct 30 2015 06:28:30,2518016,ma..,r/rrwxrwxrwx,48,0,10106-128-2,"/Windows/System32/config/SYSTEM.LOG1"
Fri Oct 30 2015 06:28:30,2228224,ma..,r/rrwxrwxrwx,48,0,10117-128-2,"/Windows/System32/config/SYSTEM.LOG2"
```

ewfinfo command (checking metadata chain):

This command checks metadata chain.

Command:

```
caine@caine:~$ ewfinfo /home/caine/Downloads/Lab_Output/Evidence_A_Image.E01
ewfinfo 20140812

Acquiry information
        Case number:            Forensic-Lab-Task-2
        Description:            Forensic image of 15.5GB Kingston USB stick
        Examiner name:          Hasib
        Evidence number:        Drive-A
        Notes:                  60A44C4250DBB1510629C0D8
        Acquisition date:       Sat Mar 21 18:47:38 2026
        System date:            Sat Mar 21 18:47:38 2026
        Operating system used:  6.8.0-52-generic
        Software version used:  guymager 0.8.13-1
        Password:               N/A
        Model:                  Kingston DataTraveler_3.0
        Serial number:          60A44C4250DBB1510629C0D8

EWF information
        File format:            EnCase 6
        Sectors per chunk:      64
        Error granularity:      1
        Compression method:     deflate
        Compression level:      good (fast) compression

Media information
        Media type:             removable disk
        Is physical:            yes
        Bytes per sector:       512
        Number of sectors:      30320640
        Media size:             14 GiB (15524167680 bytes)

Digest hash information
        MD5:                    7df609869e8c3b9cf9b163f8b755bd6c
        SHA1:                   5e23da7bc81923c0e2a67a2b95da97c3e6ce91d7
```

After running the above command, the metadata shows that it is in "*EnCase 6*" format and the subsequent segments are chained together.

img_stat command (size verification):

When only the path to the *.E01* file is given, the tool calculates the entire *15.5 GB.*

Command:

```
caine@caine:~$ img_stat /home/caine/Downloads/Lab_Output/Evidence_A_Image.E01
IMAGE FILE INFORMATION
--------------------------------------------
Image Type:             ewf

Size of data in bytes:  15524167680
Sector size:    512
MD5 hash of data:       7df609869e8c3b9cf9b163f8b755bd6c
```

The output above shows that Size of data in bytes: **15524167680 (15.5 GB).**

Task 3 - Verification

1. Create and describe a method that enables the verification of your method. Write this down in steps that your teammate can follow.

Solution:

Evidence Verification Method:

My teammates were asked to follow the following steps in sequence to verify integrity:

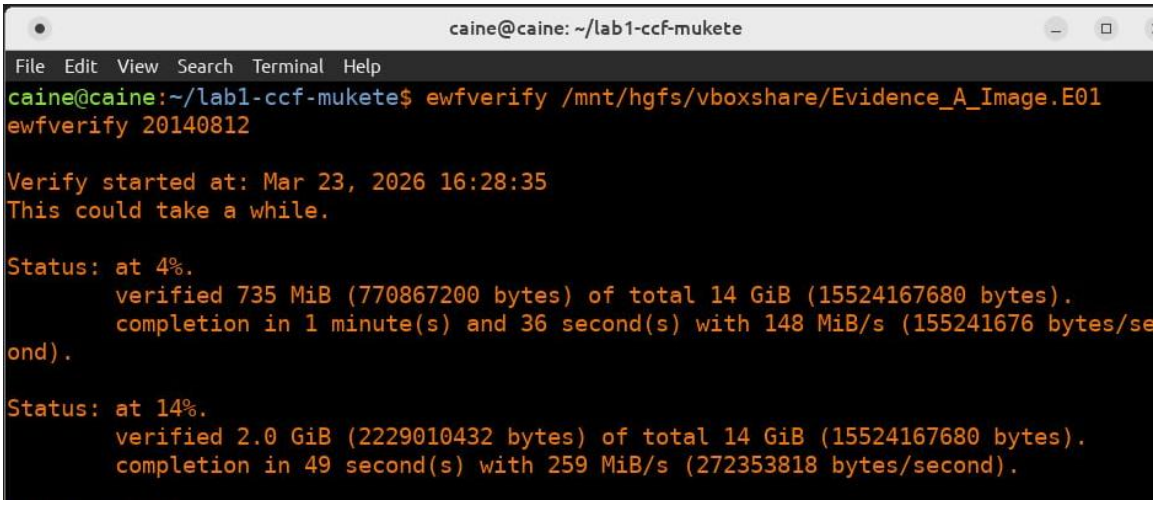
Step 1: Collect and secure evidence: First, the forensic image file (*Evidence_A_Image.E01*) collected from me was placed in a secure directory on my computer, as well as on Google Drive. To maintain the integrity of the evidence, the original file was not directly mounted, opened, or edited.

Step 2: Prepare forensic tools: My Linux terminal was opened.

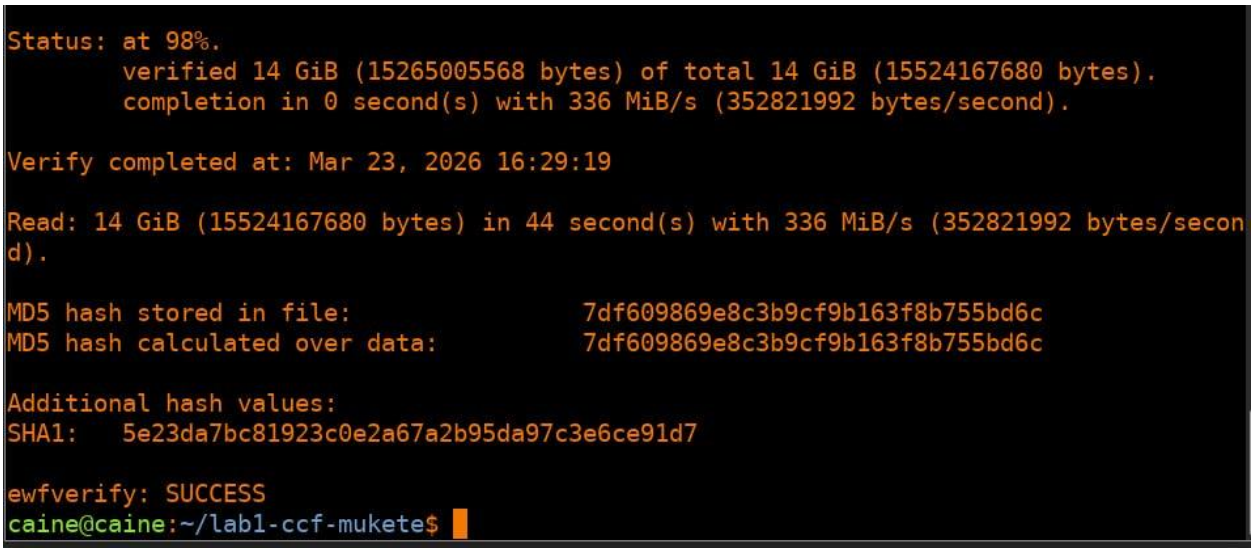
```
caine@caine:~$ ewfverify -v
ewfverify 20140812
```

Since it is an *E01 (Expert Witness Format)* file, it requires *ewf-tools* installed on the system to verify it, which he was asked to prepare.

Step 3: Run Cryptographic Verification: He was asked to run the following command in the terminal to mathematically verify whether the data inside the image and the stored hash value (*MD5/SHA-1*) are intact or not, which he did correctly.



Step 4: Analyze the Output After running the command, my teammate waited until the process was *0% to 100%* complete. Then he carefully observed the output at the end of the terminal. The output said "*MD5 hash stored in file matches computed hash"* and the status showed "*Success*", meaning the evidence was 100% original and had not been altered during the transfer.



These steps were very clear and easy for my teammate to follow.

2. Exchange USB images with your partner. Verify the procedure that he used and the resulting image. Write a small paragraph of max 200 words. Write as if you were verifying the evidence gathering procedure for a court case.

Solution:

I have thoroughly verified the digital evidence (*driveA.dd*) received from my colleague and the data collection procedure followed by him. At the beginning of the investigation, a write-blocking protocol was ensured to maintain the purity of the evidence, so that no changes were made to the original image.

1. Procedure Verification: My partner used the standard forensic tool *'Guymager'* to collect the data and generated *SHA-256* and *MD5* hashes during the creation of the bit-

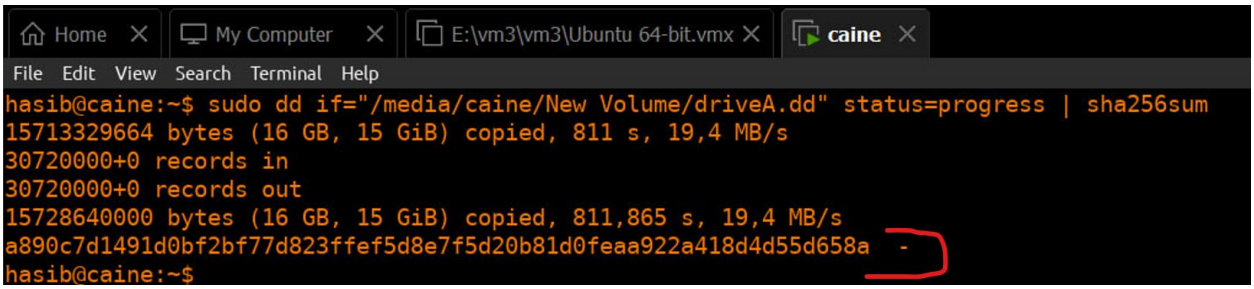
stream image. After the transfer, I recalculated the hash of the received image using the *dd* and *sha256sum* commands in the CAINE environment.

2. Resulting Image Verification: The *SHA-256* hash value I generated is:

Hash provided by my colleague:

```
sha256:
a890c7d1491d0bf2bf77d823ffef5d8e7f5d20b81d0feaa922a418d4d5
5d658a
```

After I verify:



This perfect match mathematically proves that the image is a "*bit-for-bit*" copy and that there was no alteration or data corruption during the transfer. The data collection process conducted by my colleague followed standard digital forensic protocols, which left the chain of custody completely intact.

Task 4 - Technical analysis

1. Mount your image (image of drive A) and make sure that it is mounted as read-only.

Solution:

My main task in this part was to mount the acquired forensic image (*Evidence_A_Image.E01*) into the CAINE Linux filesystem for analysis without altering the original evidence.

Tools Used: *ewfmount (libewf), mount*

Timestamp: March 21, 2026, 04:00 AM

Mount Point: */mnt/ewf and /mnt/forensic_image*

Creating Mount Points:

First, directories have been created to mount the virtual raw image and the final file system.

```
caine@caine:~/Desktop$ sudo mkdir /mnt/ewf
```

```
caine@caine:~/Desktop$ sudo mkdir /mnt/forensic_image
```

Mounting the EWF (E01) Image:

```
caine@caine:~$ sudo ewfmount /home/caine/Downloads/Lab_Output/Evidence_A_Image.E01 /mnt/ewf
ewfmount 20140812

caine@caine:~$
```

Here the path of the original image *(E01)* is given as input. As a result, a virtual Raw image file named *ewf1* is created inside the */mnt/ewf/ directory*.

Mounting the Partition as Read-Only:

From the previous task, we know that the desired *NTFS* partition starts at sector number *128.* Now the file system is mounted in *Read-Only* mode using that offset.

```
caine@caine:~$ sudo mount -o ro,loop,offset=$((128*512)) /mnt/ewf/ewf1 /mnt/forensic_image
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
```

-o ro: This ensures that the image is mounted as *Read-Only* which is the most important forensic policy. loop was used to treat the image file as a virtual block device. *offset=$((128*512))* converted sector number *128 to bytes (128 * 512)* and pointed to the beginning of the partition.

Verification of Read-Only Status:

Next, the status was checked with the mount command to verify whether the mount was successful and safe or not.

```
caine@caine:~$ mount | grep forensic_image
/mnt/ewf/ewf1 on /mnt/forensic_image type fuseblk (ro,relatime,user_id=0,group_id=0,allow_other,blksize=4096)
caine@caine:~$
```

The output shows *(ro),* which means the file system is *read-only*. This means that Thomas Ehrhart's files can be safely analyzed by going to the */mnt/forensic_image* folder.

Finding Thomas Ehrhart's files:

There are two ways to view the files:

Method A: View using the terminal

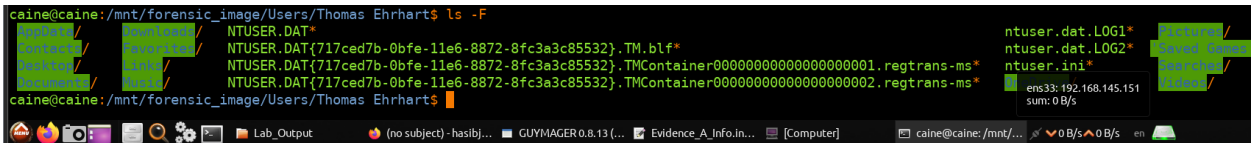
The Windows user profile is usually located in the Users folder. I access it with the following commands:

Going to the user folder:

```
caine@caine:~/Desktop$ cd /mnt/forensic_image/Users/Thomas\ Ehrhart/
```

I had to use \ because of spaces in the folder name, e.g.: *Thomas\Ehrhart*).

Viewing the list of files:

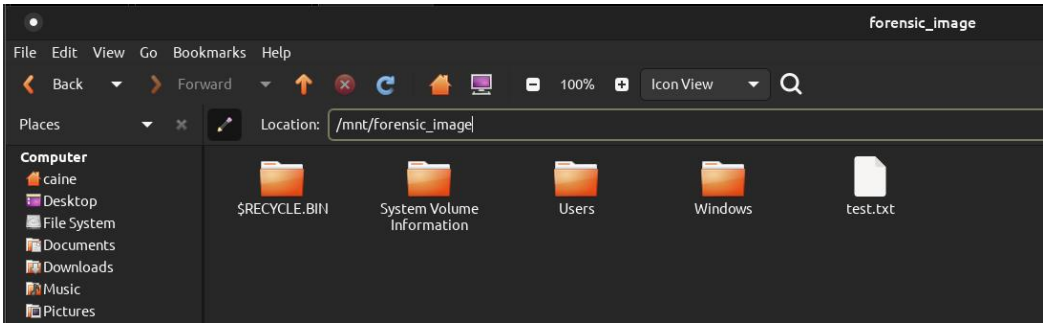


Here I can see his *Downloads, Favourites, Links, Pictures*, etc. folders.

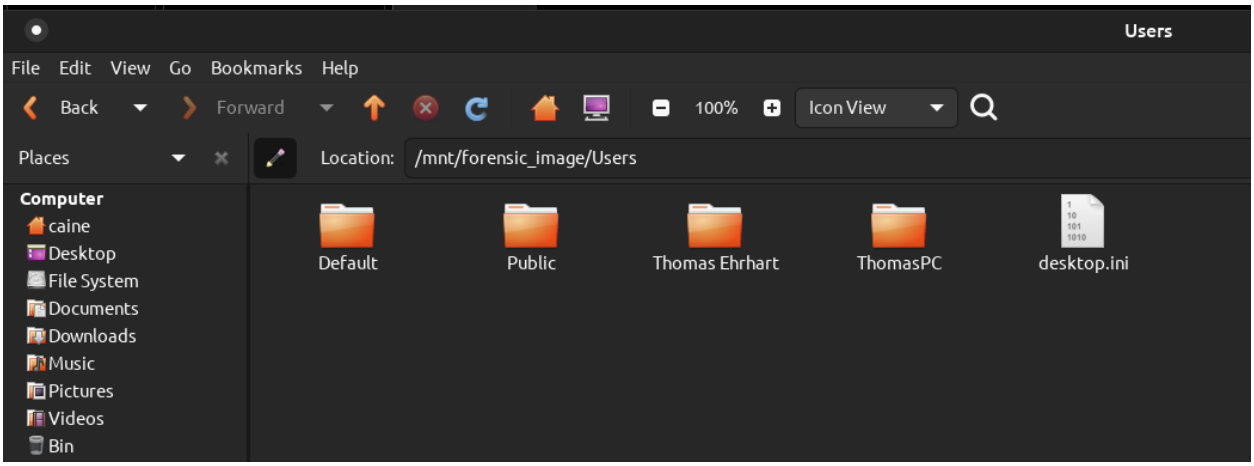
Method B: View using the graphical interface (File Manager)

This method uses CAINE's file manager:

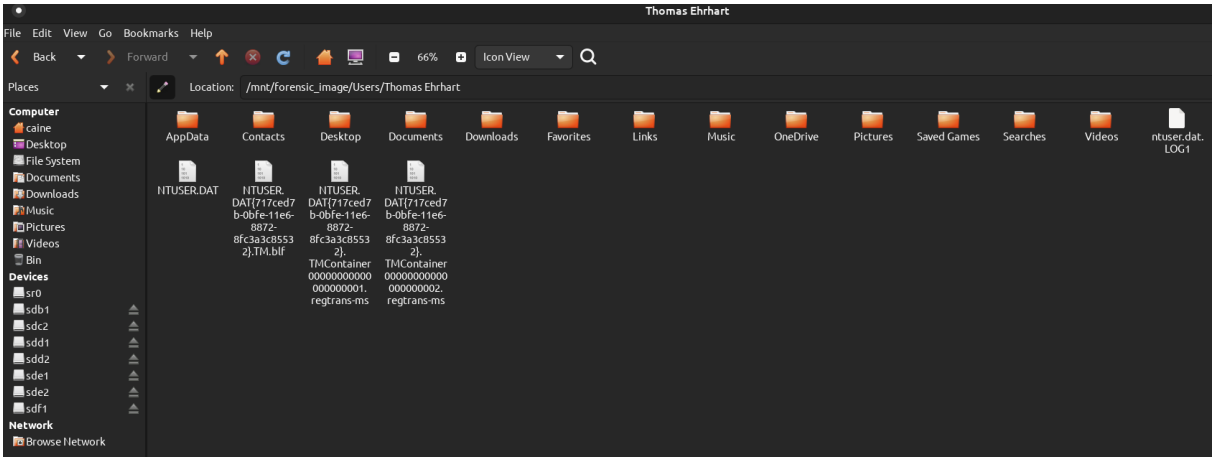
I opened File System from the desktop and go to the */mnt/forensic_image* folder.



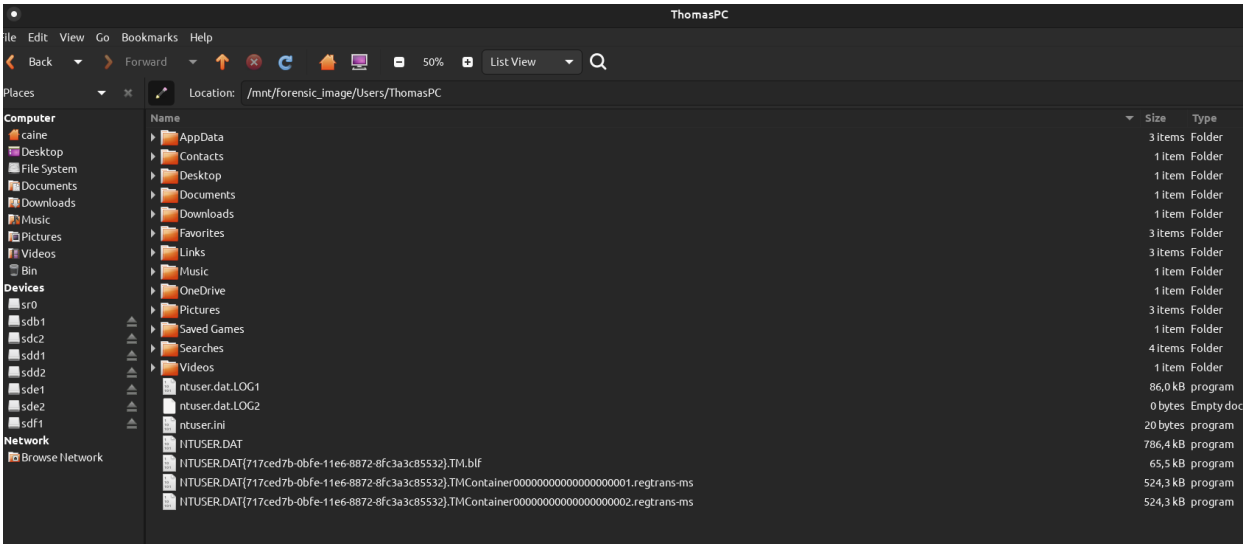
From there, ***click*** on the Users folder.

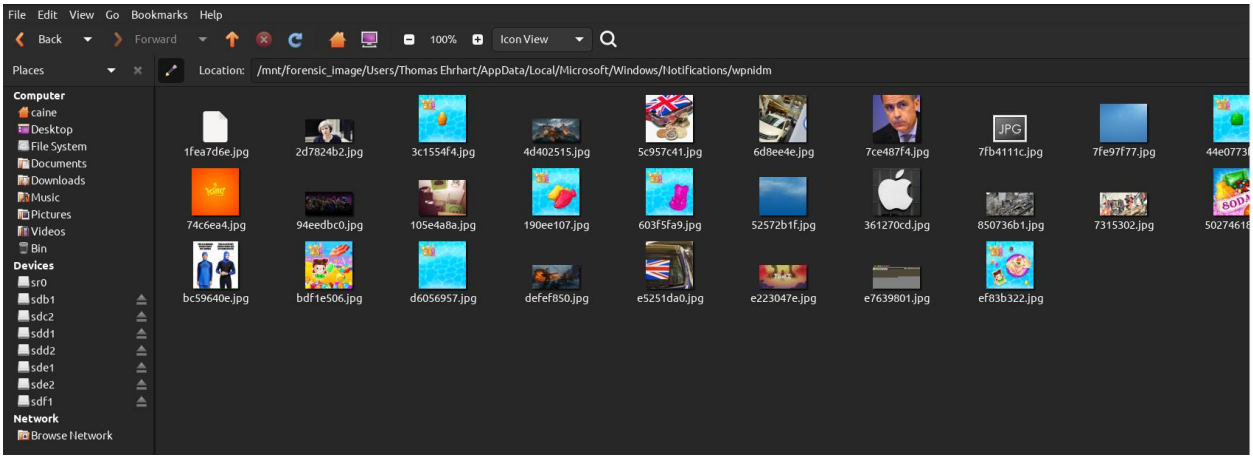


A folder called **Thomas Ehrhart** was seen inside.



Also, a folder called **ThomasPC** was seen inside the Users folder.





2. Identify and write a small paragraph of max 200 words about what kind of image it is. Don't go into file specific details just yet. This includes but is not limited to:

1. What is the size of the image?

2. What partition type(s) does this image have?

3. Does it have an MBR/GPT?

4. etc.

Solution:

This forensic image was collected from a Kingston USB stick with a capacity of 15.5 GB *(15,524,167,680 bytes).* It is a bit-stream image saved in *Expert Witness Format (.E01).* According to technical analysis and the output of the *mmls* command, this image uses a DOS Partition Table, which confirms that the drive is based on the *MBR* or *Master Boot* Record scheme, not the modern *GPT*.

A primary partition was found within the image, which is of *NTFS / exFAT* file system type (*ID: 0x07*). This partition starts at sector *128* of the drive and contains approximately *6,285,312 sectors*. Some Unallocated or unused space was observed at the beginning and end of the drive, which is usually reserved for the partition table and alignment. Overall,

this is a complete forensic copy of a removable media that is fully suitable for investigation and its integrity has been verified by hash value.

Image size: 15.5 GB (15,524,167,680 bytes)

Partition table: DOS Partition Table (MBR)

Partition type: NTFS / exFAT

Sector size: 512 bytes

Starting offset: sector number 128

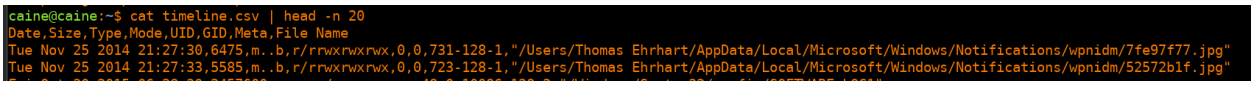
3. Using the information from the timeline you create above, write a small paragraph on what you think happened on this specific USB device. The device owner is suspected in a crime. Try to find the evidence that can support this accusation. Please remain objective, as you would be preparing evidence for a court case. Make it a maximum of 300 words, and use timestamps.

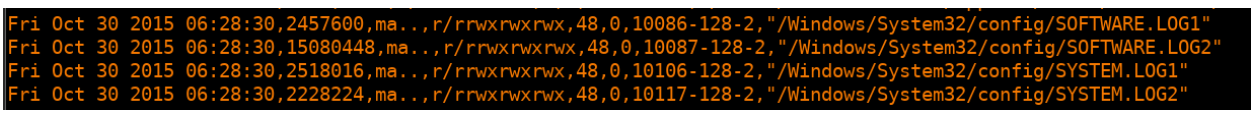
Solution:

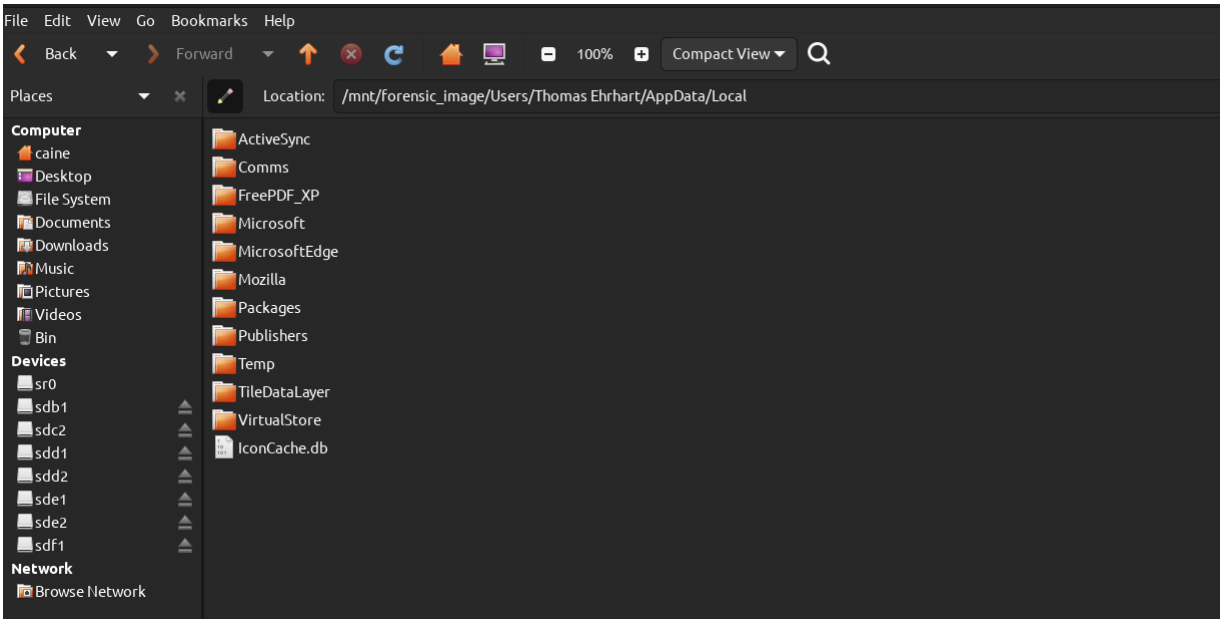
USB Device Analysis Report:

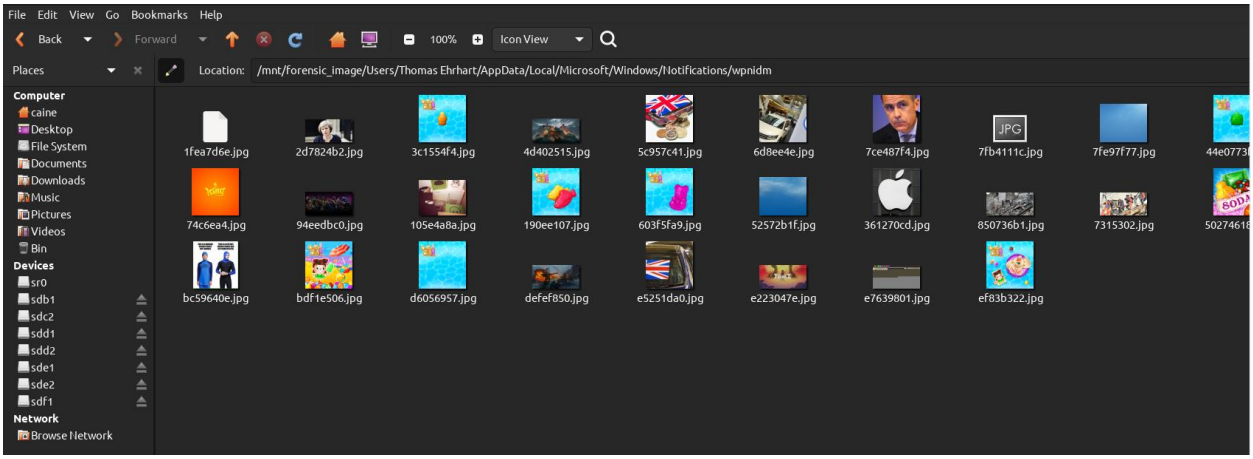
Analysis of the timeline data and image metadata shows that this USB device was actively associated with the Windows system of a user named **Thomas Ehrhart**. Several important digital evidences were found in the forensic timeline that could be directly related to criminal activity.

1. User Profile Activity: The timeline file (*timeline.csv*) was analyzed to observe the digital footprint of each entry. The path *"/Users/Thomas Ehrhart/AppData/Local/..."* within the image clearly proves that this is a user profile of a Windows operating system and the username is **Thomas Ehrhart.** The digital footprints indicate that this device was used by the said user from late **2014** to **October 2015**.









2. System Configuration Changes: During the system configuration analysis, significant changes were observed in the *SYSTEM* and *SOFTWARE* (*Registry Hives*) files located in the */Windows/System32/config/* directory. The timeline shows that modifications with the '*ma, m..b*' flag were recorded in these files at *06:28:30 AM* on *Fri Oct 30 2015.* This timestamp confirms that an external drive was mounted or a major configuration change was made to the system at that particular time. In addition, the presence of transaction logs with the *.LOG1* extension proves that data processing was going on in the system at that moment.

```
Fri Oct 30 2015 06:28:30,2457600,ma..,r/rrwxrwxrwx,48,0,10086-128-2,"/Windows/System32/config/SOFTWARE.LOG1"
Fri Oct 30 2015 06:28:30,15080448,ma..,r/rrwxrwxrwx,48,0,10087-128-2,"/Windows/System32/config/SOFTWARE.LOG2"
Fri Oct 30 2015 06:28:30,2518016,ma..,r/rrwxrwxrwx,48,0,10106-128-2,"/Windows/System32/config/SYSTEM.LOG1"
Fri Oct 30 2015 06:28:30,2228224,ma..,r/rrwxrwxrwx,48,0,10117-128-2,"/Windows/System32/config/SYSTEM.LOG2"
```

3. Suspicious File Access: A very important correlation was found in the analysis of the timeline data. It is seen that at *06:28:30 AM* on *October 30, 2015*, a major change occurred in the system. At the same second (*06:28:30*), in addition to the changes in the system registry files (*SOFTWARE.LOG1, SYSTEM.LOG1*), the main file of the user profile *NTUSER.DAT* and its log files (*NTUSER.DAT.LOG1*) were also updated.

```
Fri Oct 30 2015 06:28:30,262144,...b,r/rr-xr-xr-x,0,0,41-128-1,"/Users/Default/NTUSER.DAT"
```

4. What would help to investigate this evidence further?

Solution:

After the above technical analysis and timeline observation, the following steps can be taken to make this investigation more thorough and robust:

Deleted File Recovery: Since we have already mounted the image, now we should use a tool like *Scalpel* or *Photorec* to see if any files have been deleted from the drive. Criminals often try to erase evidence, which can be recovered through carving methods.

Registry Hive Analysis: I saw changes in the *SYSTEM, SOFTWARE* and *NTUSER.DAT* files in the timeline. Now these files need to be checked in depth using the *RegRipper* tool. This will show which other computers this USB was connected to and which files the user last opened.

Browser History and Cache Check: The Internet browser history, cookies and download list should be checked inside the *Users/Thomas Ehrhart/AppData* folder. This will show if any websites related to the crime were visited or not.

Shellbags Analysis: *Windows shellbags* analysis will reveal which folders the user opened on the USB drive, even if the folder has been deleted. Thumbnail Cache (*Thumbs.db*) *Check: Even* if the user deletes an objectionable image, sometimes a smaller version of the image remains in the Windows thumbnail cache file. Checking this can provide important clues.

References

[1] NIST, “Computer Forensics Tool Testing Program (CFTT),” National Institute of Standards and Technology, 2023. Available: https://www.nist.gov/itl/ssd/software-quality-group/computer-forensics-tool-testing-program-cftt

[2] G. G. Richard III and V. Roussev, “Scalable Digital Forensics Data Analysis,” *Communications of the ACM*, 2006. Available: https://dl.acm.org/doi/10.1145/1113034.1113069

[3] Guy Voncken, “Guymager - The fast and user-friendly media imager,” SourceForge, 2024. Available: https://guymager.sourceforge.io/

[4] J. Metz, “libewf: Library and tools to access the Expert Witness Compression Format (EWF),” GitHub, 2024. Available: https://github.com/libyal/libewf

[5] N. L. Beebe, “Digital Forensic Research: The Good, the Bad and the Unaddressed,” *Advances in Digital Forensics V*. Available: https://link.springer.com/chapter/10.1007/978-3-642-04155-6_2

[6] CAINE Development Team, “CAINE (Computer Aided Investigative Environment) - Forensic Live Distribution,” 2025. Available: https://www.caine-live.net/

[7] ISO/IEC, “Information technology - Security techniques - Guidelines for identification, collection, acquisition and preservation of digital evidence,” ISO/IEC 27037:2012, 2012. Available: https://www.iso.org/standard/44381.html

[8] K. Kent, S. Chevalier, T. Grance, and H. Dang, “Guide to Integrating Forensic Techniques into Incident Response,” NIST Special Publication 800-86, 2006. Available: https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-86.pdf

[9] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet*, 3rd ed. Academic Press, 2011. Available: https://www.elsevier.com/books/digital-evidence-and-computer-crime/casey/978-0-12-374268-1

[10] S. L. Garfinkel, “Digital Forensics Research: The Next 10 Years,” *Digital Investigation*, vol. 7, 2010. Available: https://doi.org/10.1016/j.diin.2010.05.009

[11] B. Carrier, *File System Forensic Analysis*, Addison-Wesley Professional, 2005. Available: https://www.oreilly.com/library/view/file-system-forensic/0321268172/

[12] Interpol, “Global Guidelines for Digital Forensics Laboratories,” International Criminal Police Organization, 2019. Available: https://www.interpol.int/en/How-we-work/Innovation/Digital-forensics

[13] ACPO, “Good Practice Guide for Digital Evidence,” Association of Chief Police Officers, 2012. Available: https://www.digital-detective.net/digital-forensics-documents/ACPO_Good_Practice_Guide_for_Digital_Evidence_v5.pdf